EA3798

North Coast (Areas 3 & 4) Recreational Angling Creel Survey 2017

FINAL REPORT

Prepared for:

North Coast - Skeena First Nations Stewardship Society
Prince Rupert, BC

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EXECUTIVE SUMMARY

Given its potential impact on multiple transboundary stocks of salmon and groundfish, the North Coast (Areas 3 and 4) recreational fishery must be monitored. Effort and harvest data are required for management of fisheries and monitoring of indicator stocks. Since 1997, creel surveys have been used to estimate effort and harvests, with the added benefit of the collection of biological samples such as scales, and Coded Wire Tags. The North Coast Skeena First Nations Stewardship Society (NCSFNSS) has managed this program since 2015, with support from the Pacific Salmon Commission Northern Fund. The survey consisted of access point interviews and aerial overflights, for the months of June, July and August 2017. Lodge-based recreational harvest and effort were monitored by DFO. Survey design and implementation, and analytical methods were similar to and consistent with those from the past several fishing seasons.

From 1 June to 31 August 2017, a total of 2,408 vessel interviews (involving 7,237 anglers) were collected by staff stationed at one of three access points (located in Prince Rupert or Port Edward) during 173 survey shifts. 23 aerial effort surveys were conducted. High priority was placed on gathering biosamples and fin-clip incidence rates from Chinook (*Oncorhynchus tshawytscha*) and Coho (*O. kisutch*) salmon, and length measurements from Pacific Halibut (*Hippoglossus stenolepsis*). Survey effort was adequate, as relatively good precision was achieved for of the harvest estimates for most of the main taxa (all precision goals were met).

Total fishing effort was estimated to be 13,598 vessel trips. In total, an estimated 15,416 Chinook Salmon were caught, with 10,108 fish kept and 5,308 released. It was estimated that 43,444 Coho Salmon were caught, with 38,713 fish kept and 4,731 released. Total Pacific Halibut catch was estimated to be 21,789 individuals, with 14,606 fish kept and 7,184 released. Biomass of harvested Pacific Halibut was estimated at 101.5 net tonnes (223,819 lb), including 42.2 t (93,077 lb) in Area 3 and 59.3 t (130,742 lb) in Area 4. In terms of round weight, Pacific Halibut harvest was 134.5 t (296,421 lb) including 55.9 t (123,269 lb) in Area 3 and 78.5 t (173,151 lb) in Area 4. Total harvest estimates included 2,772 Lingcod (*Ophiodon elongates*), 9,389 rockfish (*Sebastes* spp.), and 2,333 groundfish other than Pacific Halibut and Lingcod.

Creel surveyors biosampled 865 Chinook Salmon, including length measurements (median 69 cm, n = 602), adipose fin-clip incidence rates (14% were clipped; n = 864), and scale samples (n=558). For Coho Salmon, 1,918 fish were biosampled, including the measurement of fork lengths (median 64 cm, n = 1,020), and assessment of adipose fin-clip incidence rates (1.1% were clipped; n = 1,910). For 95 Chinook and 10 Coho salmon with fin clips, heads were collected for submission to the Salmon Head Recovery Program. Biosampling of Pacific Halibut included length measurements for 1,020 individuals (median = 83 cm). Average net and round weights were 7.1 kg (15.6 lb) and 9.3 kg (20.6 lb) in Area 3, and 6.9 kg (15.2 lb) and 9.1 kg (20.1 lb) in Area 4, respectively.

Recreational fishing effort, salmon CPE, and salmon catch in Areas 3 & 4 have been increasing over the last 20 years, in some cases dramatically. Pacific Halibut harvest did not show a strong among-year trend, likely the result of effective management measures that reduced daily bag limits and imposed maximum size limits. So while increasing trends highlight the need for continued monitoring of this fishery into the future, there is evidence that management measures can keep harvest under control.

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INTRODUCTION

The North Coast recreational fishery (in DFO's Pacific Region, Management Areas 3 and 4) harvests multiple species, mainly targeting Chinook (*Oncorhynchus tshawytscha*) and Coho (*O. kisutch*) salmon, but also Pacific Halibut (*Hippoglossus stenolepsis*), Lingcod (*Ophiodon elongates*), and other groundfish, rockfish (*Sebastes* spp.) and invertebrate species. Residents and tourists both enjoy sport fishing on the North Coast, and over the past 20 years effort has increased substantially (Robichaud et al. 2017). In light of increasing pressure, and of conservation concerns for some target stocks, there is a need to understand harvest from the Area 3 and 4 recreational fishery, with respect to catch rates and stock origins. And since harvested salmon can originate from BC or US (enhanced and wild) stocks, there is a critical need for this creel program to determine fishery impacts, especially for stocks of concern.

In addition to producing harvest and effort estimates with known error, this creel program has the added benefit of on-the-ground coverage, which allows for species identifications and harvest quantities to be verified, and biological samples to be collected from the recreational fishery. For example, CWT data, which are important components of the annual stock assessment data, and are required to monitor trends and manage the fisheries for Nass and Skeena river Chinook and Coho salmon stocks, come from the heads that our observers collect from randomly sampled adipose-clipped Chinook and Coho salmon. As such, the creel survey program helps the Pacific Salmon Commission develop improved information for resource management, including better stock assessment, data acquisition, and improved scientific understanding of factors that might limit salmon production.

The Area 3 and 4 creel survey program has operated during each recreational fishing season since 2008, estimating fishing effort and species-specific catch rates (Van Tongeren and Winther 2010, Robichaud et al. 2016, 2017). Starting in 2015, the responsibility for carrying out the creel survey shifted from Fisheries and Oceans Canada (Prince Rupert) to the North Coast Skeena First Nations Stewardship Society (NCSFNSS), with support from the Pacific Salmon Commission Northern Fund. NCSFNSS's creel survey followed a design similar to that used in the past by Fisheries and Oceans Canada, which, in turn, was based on an adaptation of the Strait of Georgia Creel Survey design, developed in 1982 by LGL Limited (English et al. 2002).

This report documents the 2017 recreational catch and effort statistics for fishing activities in Pacific Fishery Management Areas 3 and 4, and describes the program activities, data collection methodology and the analytical approaches that were used to generate the estimates.

SCOPE

The geographic scope of the creel survey was the same as in the previous eight years (Van Tongeren and Winther 2010, Robichaud et al. 2016, 2017), i.e., Management Areas 3 and 4 (from Portland Inlet, through Chatham Sound to Porcher Island; Figure 1). For the purposes of data collection and analysis, the study area was divided into 12 geographic strata ('subareas'), including three subareas in Area 3 (3I-3K) and nine subareas in Area 4 (4A-4H, and 4L).

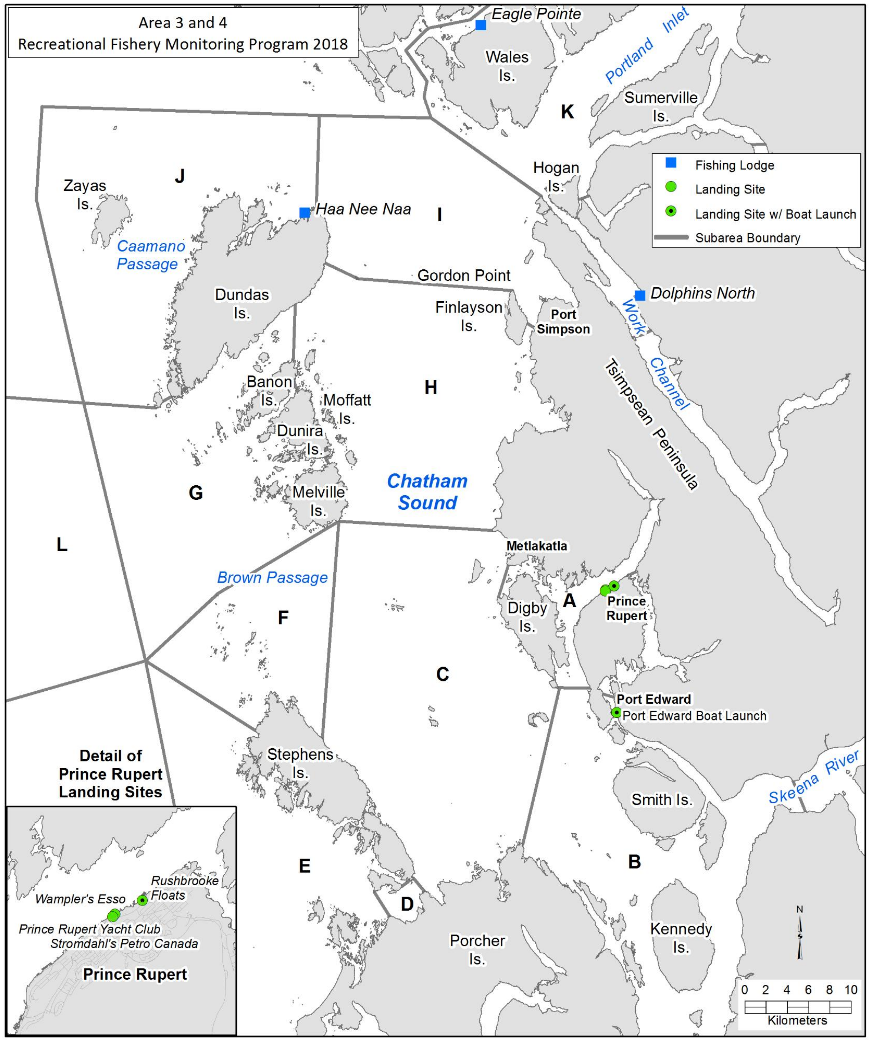

Figure 1. Map of the North Coast study area, showing the major landing sites, fishing lodges, and the boundaries of the subareas, which were used as spatial strata in the creel survey design.

The temporal scope of the full study was from June 1st through August 31st, 2017. Survey schedules were designed to provide sufficient data to derive statistically robust catch and efforts estimates for each month and subarea.

The study included creel analysis of all major local sportfish species, with greater emphasis placed on Chinook and Coho salmon, Pacific Halibut and Lingcod.

GOAL AND OBJECTIVES

The goal of this study was to conduct a creel survey of the North Coast (Areas 3 and 4) recreational fishery following the design and methodology used recently by DFO (Van Tongeren and Winther 2010), for the purpose of producing estimates of effort and catch of all species targeted by the recreational fishery. The creel survey included aerial effort counts and access point angler interviews, conducted throughout Areas 3 and 4 from June through August 2017. The precision goal for the study was to estimate total harvests of large components (such as Chinook Salmon) within 15% of the true value 19 times out of 20. Creel survey strata included temporal separation by month and day type (weekdays vs. weekend days). Spatial stratifications were based on creel survey subareas. All local fishing areas (i.e., those accessed from major boat launches in the area) were included in our survey design.

The seven objectives for this study were:

1. Continue development of software for data management and analysis comparable to DFO's CREST program (i.e., update analytical scripts which will allow incorporation of logbook data and trailer counts into the analysis);
2. Provide estimates of the catch and effort of all species targeted by the recreational fishery and associated estimates of precision;
3. Provide DFO with monthly in-season catch estimates for all salmon species and Halibut harvested in the North Coast recreational fishery;
4. Provide communication materials to recreational harvesters to convey the usefulness and importance of their catch data in fisheries management;
5. Collect biological data , such as fin clip incidence for Coho and Chinook, scale samples of Chinook Salmon, and Pacific Halibut lengths from a representative subsample of the landed harvest (during angler interviews);
6. Produce a final written report to be available to a broad audience; and
7. Present creel survey findings at the North Coast salmon post-season review meeting to a broad audience of commercial, recreational interests and First Nations.

Objective 1 was met prior to the start of the study period. Objectives 2 and 6 was met by completing this report. Objective 3 was met throughout the study period by providing timely and complete summaries of the monthly survey activities (with complete creel analyses done for each update report). Objective 4 will be met subsequent to the publication of this report, as its contents will be based on the results presented herein. Objective 5 was met by biosampling fish at access points throughout the study period: fish were measured, fin clip incidence data was collected for Coho and Chinook salmon, scales were collected from Chinook Salmon, heads were collected from adipose-clipped Chinook and Coho salmon, and all biological data were submitted to DFO at the end of study period. Objective 7 was met in late 2017.

METHODS

A complete census of catch and effort was available for lodge-based anglers. However, the remainder of the recreational fishery occurred over too large an area to completely census catch and effort. Instead, statistically robust methods were used to estimate catch by multiplying estimates of *angler effort* by estimates of *catch per unit of effort*. Separate estimates were generated for each creel monitoring subarea, month, day type (weekday vs. weekend), and species.

For each creel subarea during each month and day type, fishing effort was estimated by counting boats that were actively fishing during overflight surveys (sub-area counts were reduced by the number of lodge-based boats thought to have been visible to the aircraft). Catch per unit of effort for non-lodge boats was estimated from interviews (see data forms in Appendix 1) conducted at known access points. During interviews, anglers were asked about their catch, effort, and fishing locations. They were also asked about their hourly fishing activity patterns.

Data collected during interviews included:

- Date;
- Number of anglers in party;
- Whether or not the trip was guided by a professional;
- Subarea angler activity – for each subarea, the hours during which angling activity was conducted;
- Subarea catch – the number of fish kept and the number released by species and subarea;
- Subarea gear – the amount and type of fishing gears used to catch each species in each subarea;
- Subarea targets – the number of hours spent targeting each species in each subarea (anglers that targeted only shellfish were noted);
- Biosampling – if allowed by the anglers, harvested fish were measured and sexed. Chinook and Coho salmon were checked for adipose clips and, if clipped and permission given, heads were collected for the Salmon Head Recovery Program. Scale samples were taken from Chinook Salmon. Chinook Salmon were also examined for flesh colour;
- Refusals – whether or not the anglers refused an interview, allowed catch to be observed, or allowed catch to be biosampled; and
- Weather conditions.

The analytical methods used, the same as in Robichaud et al. (2017), provided a statistically unbiased estimate of catch per effort, assuming the anglers interviewed were representative of the entire non-lodge recreational fishery. Interviews were conducted at the three main access locations: Port Edward and Rushbrooke boat launches, and Prince Rupert Yacht Club. The locations surveyed were selected from all available access points, based on their geographical distribution and the amount of fishing activity that was conducted from that site in recent years (see Van Tongeren and Winther 2010). Wherever possible, the busiest access points were selected preferentially in order to obtain the maximum number of interviews. This approach was based on two important observations: 1) the variability in CPE (catch-per-effort) among fishing parties landing at a single access point tends to be as great as the variability in CPE among different access points within a geographic area; and 2) CPE and effort can vary substantially both within and between days at a single site (English et al. 2002). Under these conditions it is better to obtain a large number of interviews covering all temporal strata for a small number of sites than to sample a larger number of sites and obtain fewer interviews and less complete temporal coverage for any specific site. In 2017, shift frequency was increased to ensure that our goal of sampling 20% of estimated effort was met (given the intention to have “hands on” and direct observations of more fish going forward, this meant taking more time with each interview). The interview schedule was designed to capture data from representative angler parties in both day types, and over all time periods of the day. Ideally, interviews would capture data about angler activity in all creel subareas.

Interviewing sessions were to be separated into AM and PM shifts, with AM shifts occurring between 7:00 and 15:00 and PM shifts from 13:00 to 21:00. In total, 173 shifts were worked.

Fishing effort was estimated from boat-counts made during overflights. Overflights were conducted on 2-3 weekend days and usually on 5-6 weekdays each month. Timing of the overflights was targeted for peak fishing periods, and, although analytical methods allow for the occasional off-peak flight (e.g., due to weather or scheduling conflicts), all flights in 2017 coincided with peak fishing activity.

Angler Activity Patterns

Two weighting factors were used together with the interview-derived angling activity data to estimate the daily non-lodge fishing activity pattern (English et al. 2002). The first weighting factor, $W1$, expanded the numbers of days spent interviewing at each landing site to account for the total number of days available for sampling. That is, it was assumed that the daily activity pattern recorded during the interview shifts at landing site a, were consistent for landing site a, even during the days when no interviews occurred. A specific $W1$ was calculated for each landing site during each month and day type:

$$W1_{mda} = \frac{N_{md}}{K_{mda}} \qquad \text{(Equation 1)}$$

where N_{md} was the total number of type d days in month m; and K_{mda} was the number of days during which interviews occurred at landing site a, on type d days during month m.

The second weighting factor, $W2$, expanded the numbers of interviews conducted, to account for the boats that were *not* interviewed (or that refused interviews). That is, it was assumed that the activity pattern recorded during the interview shifts also held for those anglers that were not interviewed. A specific $W2$ was calculated for each surveyor (u) operating on each surveying date (k) at each landing site (a):

$$W2_{aku} = \frac{LF_{aku} + LM_{aku}\left(\dfrac{LF_{aku}}{LF_{aku} + LN_{aku}}\right)}{LF_{aku} - ri_{aku}}, \qquad \text{(Equation 2)}$$

where LF_{aku} was the number of interviewed boats that were fishing (a subset of these, ri_{aku}, refused an interview, or had an incomplete interview that could not be used), LN_{aku} was the number of interviewed boats that were not fishing, and LM_{aku} was the number of boats that were not interviewed by surveyor u, during surveying date k, at landing site a. This equation incorporates the assumption that interviewed boats were representative, allowing us to estimate the portion of missed boats that were fishing.

We used the term $A_{mdsakut}$ to denote the number of non-lodge boats reporting activity during time-block t, as interviewed by surveyor u on survey date k, at landing site a, in subarea s, during month m, and on day type d (n_{mdsaku} was used to denote the total number of boats that reported activity during *any* time-block). The two correction factors were applied, and the data were summed over surveyors, survey dates and landing sites:

$$A'_{mdst} = \sum_k \sum_u \sum_a \left(W1_{mda} \cdot W2_{aku} \cdot A_{mdsakut}\right). \qquad \text{(Equation 3)}$$

Summing the adjusted number of anglers over the hourly time-blocks gave:

$$T'_{mds} = \sum_t A'_{mdst}. \qquad \text{(Equation 4)}$$

Table 1. The amount of data (number of complete interviews) available to estimate non-lodge angler activity patterns and CPE, for all levels of each factor, 2017.

		Subarea											
Month	Day Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
June	WD	17	27	14	108	47	72	65	10	108	22	42	15
	WE	28	20	10	89	56	90	37	8	101	21	42	9
July	WD	20	32	51	131	61	112	63	10	126	34	57	18
	WE	26	21	21	102	47	102	30	7	89	28	34	5
August	WD	11	12	32	64	46	75	37	0	41	9	45	8
	WE	3	8	20	45	37	41	21	4	43	9	28	3

The proportion of non-lodge boats (P_{mdst}) that were actively fishing during each of the hourly time-blocks (t) was calculated for each month, day type, and subarea:

$$P_{mdst} = \frac{A'_{mdst}}{\sum_k \sum_u \sum_a \left(W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku}\right)} . \qquad \text{(Equation 5)}$$

Using this method, 72 unique angler activity patterns were to be estimated (i.e., 3 months × 2 day types × 12 subareas). To reliably describe angler activity, a relatively large number of boats (~60) needed to be interviewed in each of the 72 blocks. In the end, some blocks contained too few interviews (Table 1), so it was decided to pool activity data over subarea. The equation for non-lodge angler activity was thus:

$$P_{mdt} = \frac{\sum_s A'_{mdst}}{\sum_s \sum_k \sum_u \sum_a \left(W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku}\right)} , \qquad \text{(Equation 6)}$$

and the associated variance was:

$$S^2_{P_{mdt}} = \frac{(P_{mdt})(1-P_{mdt})}{\sum_s \sum_k \sum_u \sum_a \left(W1_{mda} \cdot W2_{aku} \cdot n_{mdsaku}\right)} . \qquad \text{(Equation 7)}$$

Catch Per Effort Estimation

Catch per effort (numbers of fish kept or released per boat trip) of non-lodge boats was estimated for each species from interviews of non-lodge anglers. Catch per effort was calculated separately for guided and non-guided boats, and overall; but only the overall estimates were used in subsequent calculations. For each interview (i), the guided status (g), month (m), and the catch (C) of each species (r) in each subarea (s) was recorded. Given that each interview represented one boat trip, then catch per effort (CPE), in terms of catch per boat trip, was calculated as:

$$CPE_{msrgi} = \frac{C_{msrgi}}{1} \quad \text{where } g \in \{\text{all, guided, not-guided}\}. \qquad \text{(Equation 8)}$$

Mean CPE was calculated as the average catch, over all n_{msg} boat trips:

$$C\hat{P}E_{msrg} = \frac{\sum_{i=1}^{n_{msg}} C_{msrgi}}{n_{msg}} . \qquad \text{(Equation 9)}$$

The variance for the estimate of mean catch per effort was calculated as:

$$S^2_{C\hat{P}E_{msrg}} = \frac{\sum\left(CPE_{msrgi} - C\hat{P}E_{msrg}\right)^2}{(n_{msg} - 1)}. \quad \text{(Equation 10)}$$

In the event that a month/subarea-specific sample size was < 5, adjacent subareas would be pooled together as follows:

- if n for 3I, 3J or 3K was too low, then it was combined with the other two to give a 3IJK regional average CPE;
- if n for 4A, 4B or 4C was too low, then it was combined with the other two to give a 4ABC regional average CPE;
- if n for 4D, 4E or 4F was too low, then it was combined with the other two to give a 4DEF regional average CPE; and

if n for 4G, 4H or 4L was too low, then it was combined with the other two to give a 4GHL regional average CPE.

Similar methods were used to calculate the average number of fish harvested (retained) per boat trip (HPE), and the average number of fish released per boat trip.

Angler Effort Estimation

To obtain statistically valid estimates of non-lodge fishing effort, boats were counted during overflight surveys conducted from a fixed-wing aircraft traveling along a set path over the study area (see Figure 2). In order to calculate the non-lodge effort, the estimated counts of active lodge boats needed to be subtracted from the overflight counts for that particular day.

Lodge Effort Census

A complete census of lodge-based angling effort was collected from lodge log books by DFO personnel (I. Winther, DFO Prince Rupert, pers. comm.). DFO used the following procedure to estimate the number of active lodge vessels that would have been available to be observed during the flights: 1) Lodge based vessels were considered unavailable to the flights on change-over days (days when the lodges received new clients; on typical change-over days, departing clients leave before the flight and new clients arrive after the flight). Change-over days occurred twice a week but timing was often different for each lodge, making 29% (2/7) of the lodge effort not visible to flights due to change-over days. 2) Lodges reported their lunchtime activity to DFO, which ranged from virtually always going in for a mid-day break to rarely leaving the water during the day. Thus, approximately 60% of the remaining vessel activity was available to the

Table 2. Summary of overflight sampling effort (number of flights shifts by month and day type), 2017.

Month	*Weekday*	*Weekend/ Holiday*	*Total*
June	5	3	8
July	5	3	8
August	5	2	7
Total	15	8	23

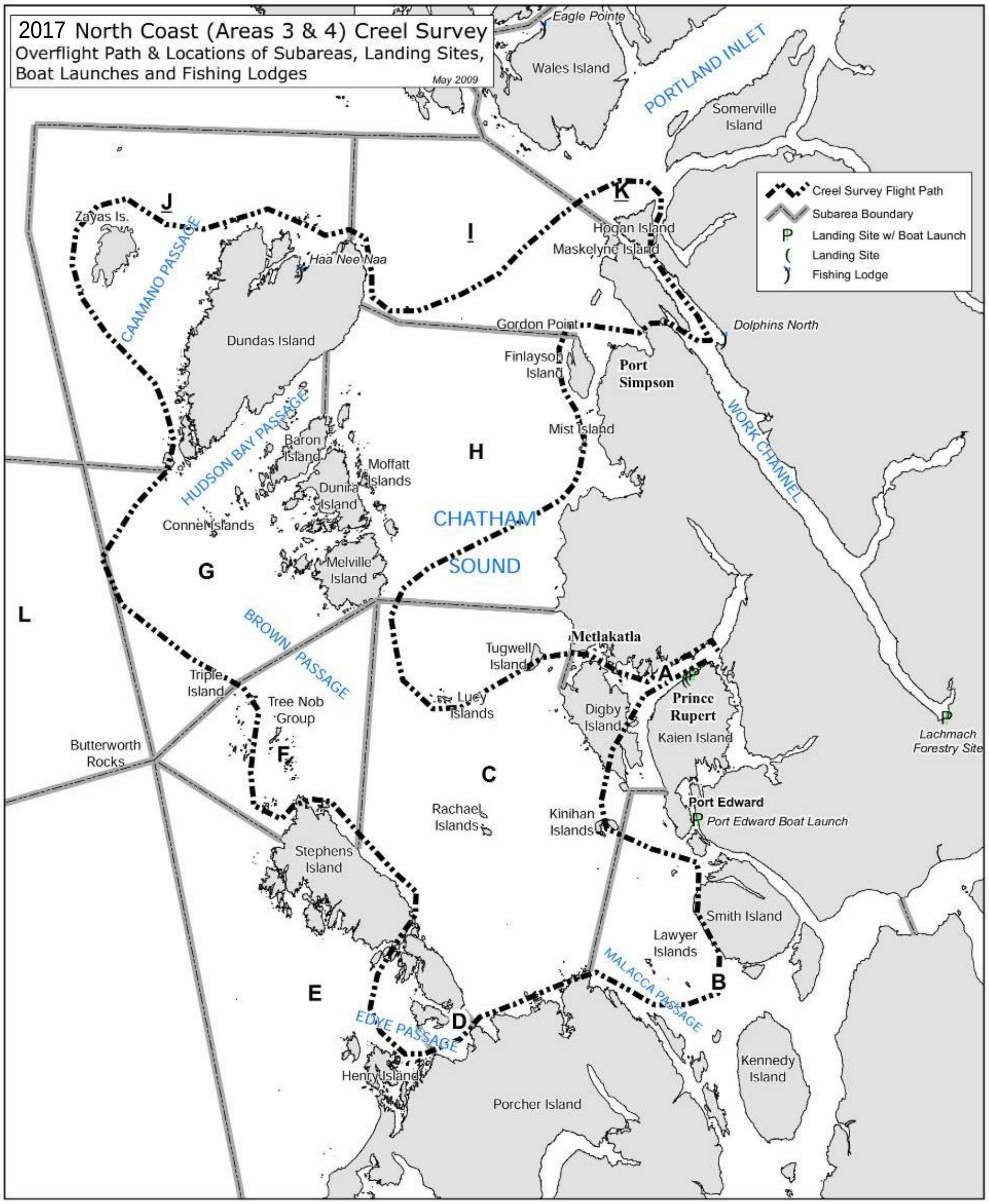

Figure 2. Map showing planned flight path of effort survey overflights. Letters indicate subareas. Flights originated and ended in Prince Rupert, and flew in the clockwise direction. On good weather days, the flights also included a foray into subarea 4L (path not shown).

flight counts on 'regular' days (i.e., not change-over). 3) Vessels on mid-day break or on change-over days were excluded from the numbers of vessels visible to the flights. All of the lodge catch was reported from Area 3, without subarea-specific effort information. Thus, lodge effort was partitioned among the Area 3 subareas, see below.

Non-lodge Effort Surveys

Table 2 shows the number of overflight surveys scheduled for each month and day type. Each survey was designed to allow visual inspection of the subareas through which the aircraft passed.

Each time the aircraft entered a subarea, the start and end times were recorded, along with the numbers of small (< 30 ft) and large (≥ 30 ft) recreational vessels that were observed, tallied separately depending on whether or not they were actively fishing at the time.

During survey *o* (conducted during month *m* and on day type *d*), observers tallied the total number of recreational boats that were actively fishing at time *t* on their i^{th} pass over subarea *s*, V_{mdsoit}. For each visit to a subarea within Management Area 3, the relative proportion of boats was calculated, ρ_{mdsoit}, and the total number of lodge boats (λ_{mdo}) apportioned accordingly:

$$\lambda_{mdsoit} = \rho_{mdsoit} \cdot \lambda_{mdo} \quad . \qquad \text{(Equation 11)}$$

The number of non-lodge boats was hence:

$$v_{mdsoit} = V_{mdsoit} - \lambda_{mdsoit} \, . \qquad \text{(Equation 12)}$$

Since angling occurs over the course of the entire day, the number of non-lodge boats that were observed was divided by the proportion of average daily number of boats active (P_{mdst}) during the time block when the observations were recorded. Since the flight path (Figure 2) brought the observers into some subareas more than once during the survey (covering different ground, thus the tallies were considered additive), the adjusted tallies were summed over all visits, to calculate the total number of active non-lodge fishing boats on the day of the survey, by subarea, B_{mdso}:

$$B_{mdso} = \sum_{i} \frac{v_{mdsoit}}{P_{mdst}} \quad . \qquad \text{(Equation 13)}$$

The variance of B_{mdso} was calculated using the standard formulas for combining the variance of products and quotients of two independent random variables (Goodman 1960):

$$\begin{aligned} &\text{if } z = x/y,\ Var(z) = (y^{-2})Var(x) + (x^2 y^{-4})Var(y) \ ; \\ &\text{if } z = xy,\ Var(z) = (y^2)Var(x) + (x^2)Var(y) \ ; \text{ and} \qquad \text{(Equation 14)} \\ &\text{if } z = cx,\ Var(z) = (c^2)Var(x) \quad \text{where c is a constant.} \end{aligned}$$

Since there was no variance associated with the boat counts,

$$S^2_{B_{mdso}} = (v_{mdsoit})^2 (P_{mdst})^{-4} S^2_{P_{mdst}} \, . \qquad \text{(Equation 15)}$$

Next, the survey-specific 'daily effort' estimates were averaged over the number of surveys conducted, n_{mds}, as:

$$\hat{B}_{mds} = \frac{\sum_{o=1}^{n_{mds}} B_{mdso}}{n_{mds}} \ , \qquad \text{(Equation 16)}$$

with a variance of:

$$S^2_{\hat{B}_{mds}} = (n_{mds})^{-2} S^2_{B_{mdso}} \, . \qquad \text{(Equation 17)}$$

Total monthly fishing effort, was calculated for each day type and subarea by multiplying the average daily effort by the number days of day type *d* that occurred in month *m*:

$$E_{mds} = \hat{B}_{mds} \cdot N_{md} \, . \qquad \text{(Equation 18)}$$

with a variance of:

$$S^2_{E_{mds}} = S^2_{\hat{B}_{mds}} \cdot N^2_{md} \qquad \text{(Equation 19)}$$

The standard error of the estimate of the total monthly fishing effort, after pooling over day types, was:

$$S_{E_{ms}} = \sqrt{\frac{\sum_d S^2_{E_{mds}}}{\sum_d n_{mds}}}. \quad \text{(Equation 20)}$$

Because of inclement weather, not all overflights surveyed every subarea. Especially for the more exposed subareas, it was possible to have no overflight data in a given combination of month and day type. In this event, subarea-specific effort was scaled using: 1) the proportion of effort in that subarea, relative to that in a three-subarea cluster (see pooling of adjacent subareas for CPE estimation, above), measured during overflights for which all three subareas were surveyed; and 2) the effort in the cluster's other two subareas, estimated for the combination of month and day type during which the data gap occurred.

Catch and Harvest Estimation

Total catch was calculated for each month, subarea and species by multiplying total angling effort by catch per effort, and then summing over day type:

$$C_{msr} = \sum_d \left(E_{mds} \cdot C\hat{P}E_{msrg} \right) \quad \text{(Equation 21)}$$

where g = 'all' (see Equation 8). The standard errors for these catch estimates were:

$$S_{C_{msr}} = \sqrt{\sum_d \left(E^2_{mds} \frac{S^2_{CPE_{msrg}}}{n_{msg}} + CPE^2_{msrg} \frac{S^2_{E_{mds}}}{n_{mds}} + \frac{S^2_{CPE_{msrg}}}{n_{msg}} \frac{S^2_{E_{mds}}}{n_{mds}} \right)} \quad \text{(Equation 22)}$$

where n_{mds} and n_{msg} are the sample sizes for E and CPE estimates, respectively.

Similar methods were used to estimate monthly numbers of fish harvested, and monthly numbers of fish released.

Biosampling

When given permission by the anglers, observers collected biological data from Chinook and Coho salmon and Pacific Halibut. All three species were sexed. Chinook and Coho salmon were examined for clipped adipose fins (mark rate data used by DFO for coded wire tag analyses) and potentially head tags. The flesh color was noted for Chinook Salmon. Scale samples were collected from Chinook Salmon (used by DFO to determine age, and stock composition).

Ideally, all anglers would allow their catch to be biosampled, and that the examined fish were an unbiased sample of all harvested fish. But Robichaud et al. (2016) theorized that the interview protocol, used until the end of 2015, may have resulted in bias. The protocol had observers ask anglers about adipose clips before asking permission to observe the fish. Anglers were often engaged in a conversation about the purpose of the fin-clips and how the data would be used, and it appeared that anglers with clipped fish were more likely to permit handling and biosampling of catch than when no clipped fish were present. To avoid this bias, we modified our data collection protocol in 2016 so that anglers were asked if their catch could be biosampled before any other interactions took place. Also starting 2016, no fin clip data were accepted on the word of the anglers: the only fin-clip data we used were those which came from observers when they handled and inspected the catch.

Biomass of Harvested Pacific Halibut

Halibut lengths were converted to weights using the International Pacific Halibut Commission (Clark 1992) regression equations:

$$\begin{aligned}\text{Weight(lb)}_{\text{net}} &= 6.921\times10^{-6} \times \text{FL(cm)}^{3.24}\\ \text{Weight(lb)}_{\text{round}} &= 9.166\times10^{-6} \times \text{FL(cm)}^{3.24}\end{aligned} \qquad \text{(Equation 23)}$$

Converting average length to average weight allowed harvested biomass to be calculated from harvested fish counts.

Statistical Analyses

For each fish taxa, the effects of month and day-type on CPE, effort, catch and harvest were examined using Kruskal Wallis tests (Zar 1984). When statistically significant effects were observed, *post-hoc* Kruskal-Wallis tests were used to compare between each possible pair, where the false discovery rate was controlled using the Benjamini–Hochberg procedure (Benjamini and Hochberg 1995).

Trailer Census

Daily between noon and 2 PM (i.e., similar to the timing of aerial overflights), the number of parked trailers at the landing location was recorded. Theoretically, trailer counts could be used as an index of fishing effort, and are more economical than overflights and are not weather dependant, though no sub-area-specific effort data is collected. DFO has been collecting trailer count data for years and we are maintaining the time series.

Linear regression was used to determine the relationship between trailer counts and fishing effort. A data point was generated each time a trailer census occurred on the same day as an overflight. The location-specific trailer counts and non-lodge boat counts were used as the independent and dependent variables, respectively.

RESULTS

Angler Interviews

Over the three month study period, there were 173 observer shifts, during which 2,408 interviews were conducted, involving 7,237 non-lodge anglers (Table 3; Figure 3). July was the busiest month, with 1051 interviews conducted (average 16.7 interviews per shift), which represented 44% of the total number of study period interviews (Table 3). In all there were 64 refusals, or 2.7% refusal rate. The highest refusal rate (5.0%) was at Rushbrooke Boat Launch, where 83% of the refusals occurred. The refusal rates for Prince Rupert Yacht Club and Port Edward Boat Launch were 0.3% and 1.2%, respectively.

Table 3. Monthly number of shifts numbers of boats interviewed per landing site, 2017. Non-lodge effort is expressed in units of boat trips. Sampling efficiency is also calculated.

	June				July				August				TOTAL			
Landing Site	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint	# ints	# anglers	# stints	ints/stint
Port Edward Boat Launch	266	712	19	14.0	301	877	21	14.3	181	527	17	10.6	748	2116	57	13.1
Prince Rupert Yacht Club	228	742	20	11.4	253	819	19	13.3	115	394	18	6.4	596	1955	57	10.5
Rushbrooke Boat Launch	321	937	19	16.9	497	1509	23	21.6	246	720	17	14.5	1064	3166	59	18.0
TOTAL	815	2,391	58	14.1	1,051	3,205	63	16.7	542	1,641	52	10.4	2,408	7,237	173	13.9
% of Total Interviews	34%				44%				23%							
Estimated Non-lodge Effort	3,925				5,389				2,424				11,738			
% of Non-lodge Effort	33%				46%				21%							
% of Effort Sampled	21%				20%				22%				21%			

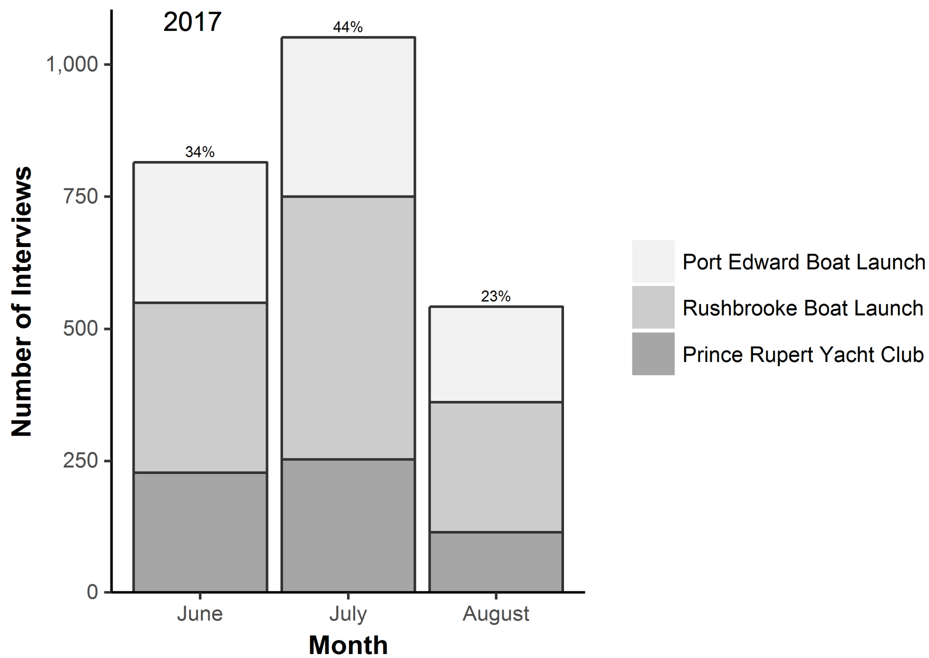

Figure 3. Monthly number of interviews conducted at each of three landing sites, 2017.

Table 4. Numbers of guided (i.e., led by professional guides) and unguided boat trips, by landing site, 2017.

	Trip Type		
Landing Site	*Guided*	*Unguided*	*% guided*
Port Edward Boat Launch	18	730	2.4%
Prince Rupert Yacht Club	273	323	46%
Rushbrooke Boat Launch	276	788	26%
TOTAL	567	1,841	24%

On average, 24% of the non-lodge boat trips were guided by a professional. The guiding rate varied by landing site. Only 18 of 748 trips interviewed at Port Edward were guided, whereas trips departing from Rushbrooke and Prince Rupert Yacht Club were guided 26% and 46% of the time, respectively (Table 4).

Non-lodge boat trip activity was unevenly distributed among the subareas, with 4C and 4F being the overwhelming favourites (Table 5). Subareas 4A, and 4B were the next most likely areas for anglers to have reported activity. The subareas with the lowest number of trips reporting activity were 4E and 4L. Only 86 interviews were of parties that only targeted shellfish, and those were largely in subarea 4A (Table 6).

Angler Activity Patterns

Monthly non-lodge angler activity patterns estimates are shown in Figure 4. There was virtually no difference between the weekday and weekend/holiday activity patterns for June or July. In August, AM activity was lower on weekdays than on weekends. Angler activity patterns showed very little change over the three-month survey period. Timing of our overflights lined up with peak activity levels in all months.

Catch Per Effort Estimates

Sample sizes were adequate (n >=5) to produce reliable estimates of non-lodge CPE in most combinations of months and management areas, except in subarea 4E in August (Table 7). However, when the interview data were split between guided and non-guided groups, monthly sample sizes did not suffice in some of the management areas (e.g., 4D-E, and 4L) where guiding was less common (Table 7). In those cases where sample sizes were limited, data from three adjacent subareas were pooled together for CPE calculations (Table 8). Average number of fish harvested per non-lodge boat trip (HPE) was calculated for each species by month, subarea, and by trip type (Appendix 2).

HPE varied significantly among months (Table 9) for Chinook ($\chi^2 = 7.7$, df = 2, $P = 0.022$), Coho ($\chi^2 = 16.9$, df = 2, $P < 0.001$), and Pink salmon (*O. gorbuscha*, $\chi^2 = 16.1$, df = 2, $P < 0.001$). For Chinook Salmon, median HPE in June was significantly higher than in August (Table 9; Figure 5). For Coho and Pink salmon, HPE in June was significantly lower than in July or August (Table 9; Figure 5). There were no significant differences among months for

HPE of Chum (*O. keta*) or Sockeye salmon (*O. nerka*), or for any species of groundfish (including Pacific Halibut and Lingcod; Figure 5). For Yellowtail Rockfish (*Sebastes flavidus*),

Table 5. Fishing locations as reported by interviewed non-lodge parties, by month and subarea, 2017. For interviews that reported activity in multiple subareas, the boat trip tally was evenly partitioned among the subareas visited.

	Subarea															
Month	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*	*3?*	*4?*	*Refusal*	*TOTAL*
June	33	34	20	114	87	122	85	14	154	30	64	13	5	32	9	815
July	40	39	58	142	93	175	83	16	174	53	70	15	14	46	34	1,051
August	11	14	37	67	69	99	48	4	71	15	57	8	8	17	21	542
TOTAL	83	87	114	323	249	395	215	33	399	98	192	36	26	95	64	2,408

Table 6. Numbers of non-lodge boat trips that reported only fishing for shellfish, by month and subarea, 2017.

	Subarea															
Month	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*	*3?*	*4?*	*Refusal*	*TOTAL*
June	0	0	0	22	3	0	0	0	0	0	0	0	0	1	0	26
July	1	0	0	35	2	1	0	0	0	1	0	0	0	0	0	40
August	0	0	0	18	1	1	0	0	0	0	0	0	0	0	0	20
TOTAL	1	0	0	75	6	2	0	0	0	1	0	0	0	1	0	86

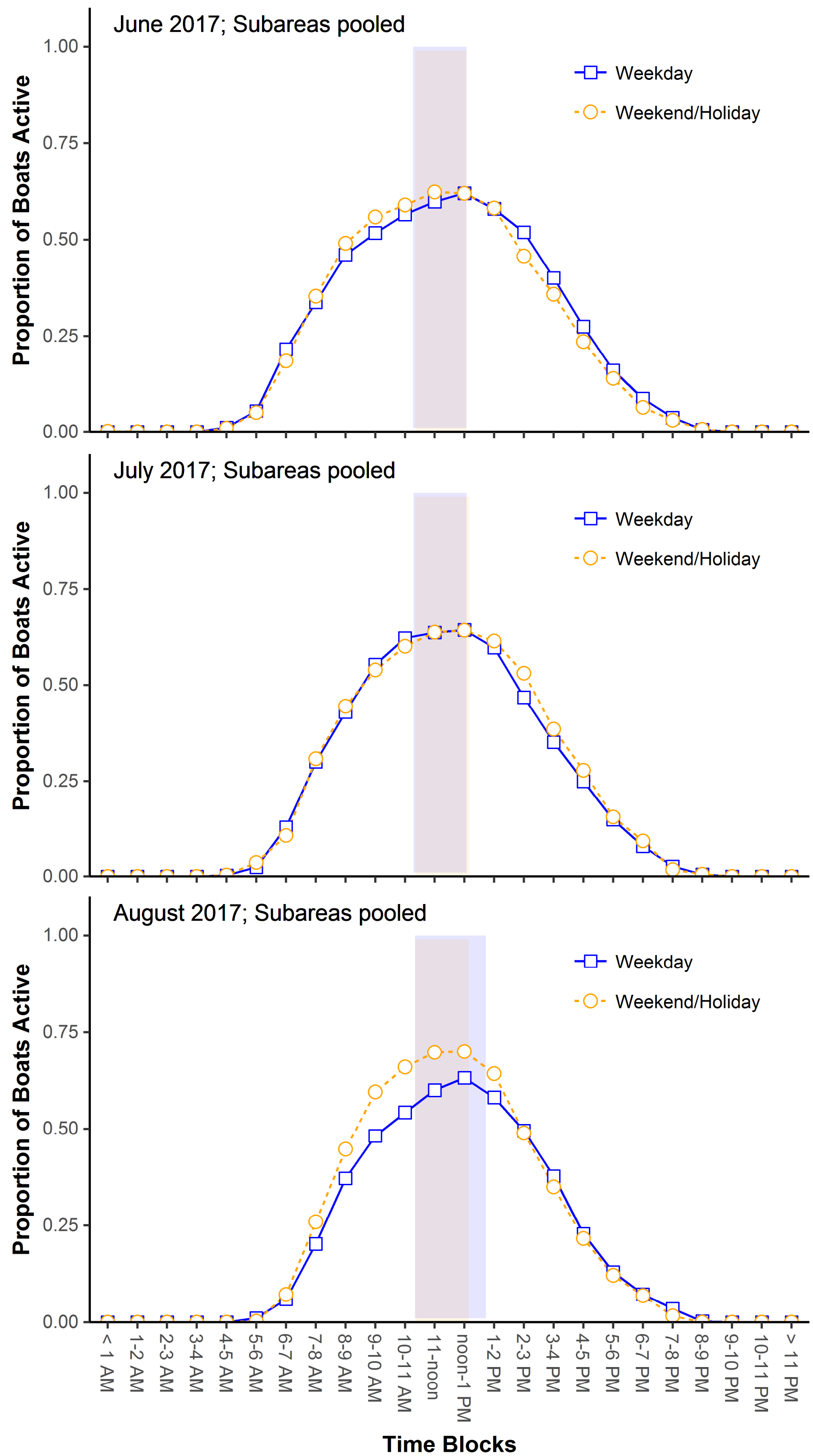

Figure 4. **Angler activity pattern (proportion recreational angling boats that were actively fishing during each hour-long time block) by day type and month, pooled over all subareas, 2017. Shaded regions bound the earliest and latest flight times for each day type (blue = weekday; orange = weekend/holiday) and month.**

Table 7. Sample sizes for CPE (harvest per effort) estimates by month, for guided and not guided non-lodge boat trips, 2017.

		Subarea											
Month	*Trip Type*	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
June	*Guided*	15	21	5	89	6	13	18	5	75	17	31	23
	Unguided	30	26	19	108	97	149	84	13	134	26	53	1
	Combined	45	47	24	197	103	162	102	18	209	43	84	24
July	*Guided*	13	16	40	69	8	12	2	2	65	12	35	14
	Unguided	33	37	32	164	100	201	91	15	149	50	55	9
	Combined	46	53	72	233	108	213	93	17	214	62	90	23
August	*Guided*	5	11	25	37	6	7	1	0	21	5	34	9
	Unguided	9	9	27	72	77	109	57	4	63	13	39	2
	Combined	14	20	52	109	83	116	58	4	84	18	73	11

Table 8. Subarea data (pooled over several adjacent areas, in some cases) used to calculate CPE in each month and subarea, for guided and not guided non-lodge boat trips, 2017.

		Subarea											
Month	*Trip Type*	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
June	*Guided*	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4G,H,L
	Combined	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
July	*Guided*	3I	3J	3K	4A	4B	4C	4D,E,F	4D,E,F	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
	Combined	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
August	*Guided*	3I	3J	3K	4A	4B	4C	4D,E,F	4D,E,F	4F	4G	4H	4L
	Unguided	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4G,H,L
	Combined	3I	3J	3K	4A	4B	4C	4D	4D,E,F	4F	4G	4H	4L

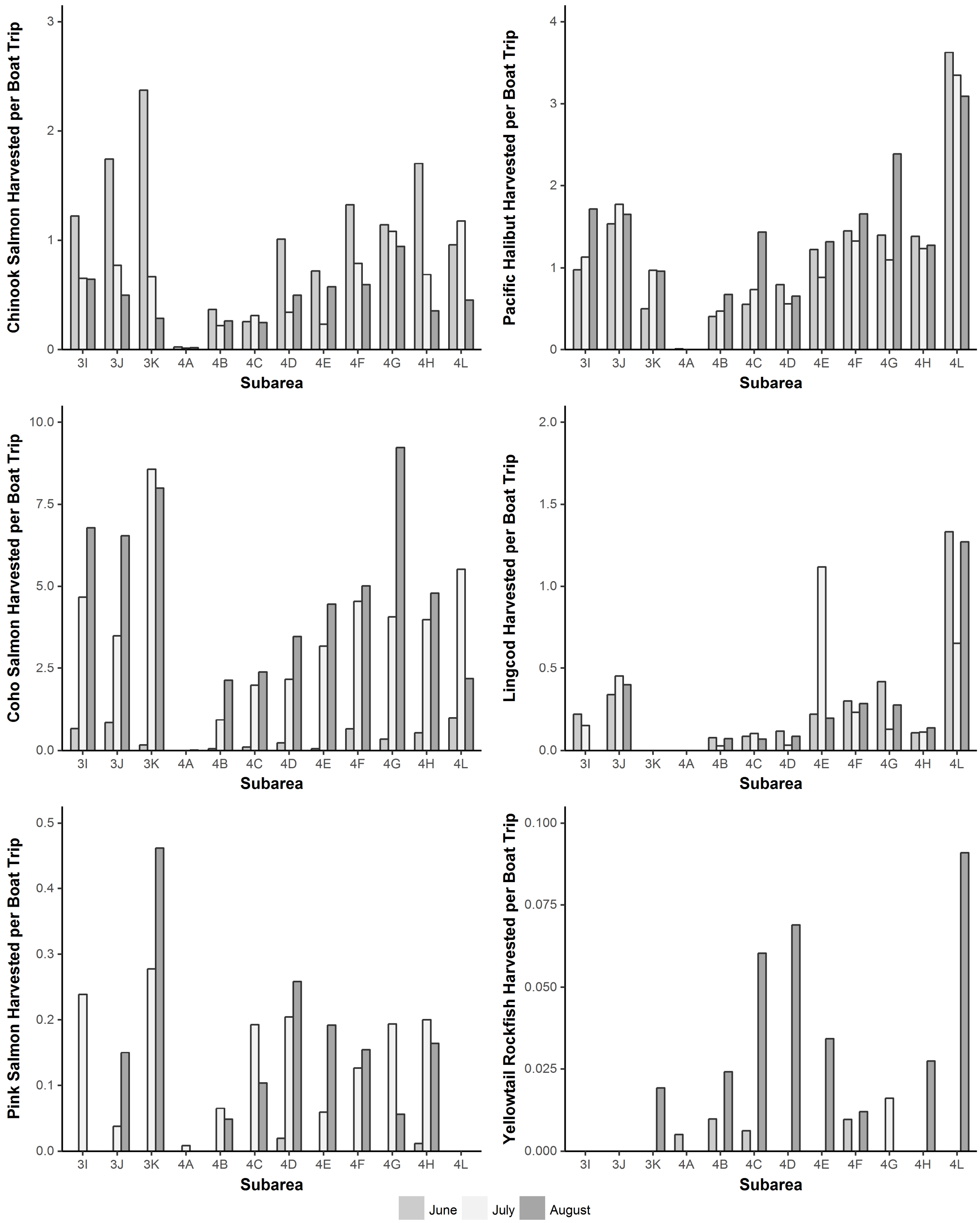

Figure 5. Average number of fish harvested per non-lodge boat trip, for six sportfish species, by month and subarea, 2017. Data are for guided and unguided trips combined. Note that Y axis scale varies among plots.

Table 9. Kruskal-Wallis tests for differences in HPE (fish harvested per non-lodge boat trip) among months, by species, 2017. Subareas treated as replicates.

	Kruskal-Wallis Statisitics			Median HPE		
Species	χ^2	df	P	June	July	August
Chinook Salmon	7.7	2	0.022	1.1	0.7	0.5
Chum Salmon	1.5	2	0.47	0	0	0
Coho Salmon	16.9	2	0.0002	0.3	3.7	4.6
Pink Salmon	16.1	2	0.0003	0	0.2	0.1
Sockeye Salmon	0.1	2	0.93	0	0	0
Arrow Flounder	0.4	2	0.80	0	0	0
Big Skate	5.5	2	0.06	0	0	0
Cabezon	2.0	2	0.37	0	0	0
English Sole	4.1	2	0.13	0	0	0
Greenling	0.8	2	0.65	0	0.002	0.004
Lingcod	0.4	2	0.81	0.2	0.1	0.1
Pacific Cod	1.2	2	0.55	0.01	0.01	0
Pacific Halibut	1.7	2	0.43	1.1	1.0	1.4
Ratfish	1.0	2	0.60	0	0	0
Rock Sole	3.3	2	0.19	0	0	0
Sablefish	0.3	2	0.84	0.09	0.08	0.07
Starry Flounder	0.6	2	0.72	0	0	0
Black Rockfish	5.8	2	0.06	0.010	0.08	0.1
Boccacio Rockfish	3.1	2	0.22	0	0	0
Canary Rockfish	1.2	2	0.55	0	0	0
China Rockfish	1.0	2	0.61	0	0	0
Copper Rockfish	1.0	2	0.59	0.02	0.008	0.03
Dusky Rockfish	3.2	2	0.20	0	0	0
Quillback Rockfish	3.7	2	0.16	0.2	0.3	0.2
Redband Rockfish	3.7	2	0.16	0	0	0
Redstripe Rockfish	3.6	2	0.17	0	0	0
Tiger Rockfish	1.5	2	0.47	0	0	0
Widow Rockfish	2.0	2	0.37	0	0	0
Yelloweye Rockfish	1.4	2	0.49	0.1	0.09	0.2
Yellowtail Rockfish	11.7	2	0.003	0	0	0.02

HPE in June was significantly lower than in July or August (χ^2 = 11.7, df = 2, P = 0.003; Table 9; Figure 5). No other rockfish species showed significant differences among months in the number of fish harvested per boat trip.

Guided trips had significantly higher HPE than non-guided trips (Table 10) for Chinook (χ^2 = 12.2, df = 1, P = 0.0005) and Coho salmon (χ^2 = 5.9, df = 1, P = 0.015), Lingcod (χ^2 = 4.4, df = 1, P = 0.036), and Pacific Halibut (χ^2 = 13.2, df = 1, P = 0.0003), among other species. Guided trips caught 240-303% more salmon, 345% more Lingcod, and 168% more Pacific Halibut than unguided trips.

For lodge boat trips, CPE estimates are shown in Table 11. On average, lodge boats retained fewer Chinook Salmon, Coho salmon, and Lingcod when compared to guided non-lodge boat trips (76%, 36%, and 81% of the guided non-lodge boats, respectively), but did better than non-guided non-lodge boats (183%, 108%, and 280%). Whereas lodge boats retained fewer Pacific Halibut (average 0.5 fish per trip) than either guided (1.7) or non-guided (1.0) non-lodge boats (Tables 10 and 11).

Table 10. Kruskal-Wallis tests for differences in HPE (fish caught per non-lodge boat trip) between guided and non-guided boats, by species, 2017. Months and subareas treated as replicates.

	Kruskal-Wallis Statisitics			Median HPE	
Species	χ^2	df	P	Guided	Not
Chinook Salmon	12.2	1	0.0005	1.1	0.4
Chum Salmon	1.0	1	0.31	0	0
Coho Salmon	5.9	1	0.015	6.3	2.1
Pink Salmon	0.8	1	0.36	0	0.05
Sockeye Salmon	0.1	1	0.81	0	0
Arrow Flounder	0.3	1	0.56	0	0
Big Skate	10.1	1	0.001	0	0
Cabezon	1.0	1	0.32	0	0
English Sole	2.0	1	0.15	0	0
Greenling	8.6	1	0.003	0	0
Lingcod	4.4	1	0.036	0.3	0.10
Pacific Cod	0.1	1	0.71	0	0
Pacific Halibut	13.2	1	0.0003	1.7	1.0
Ratfish	2.0	1	0.15	0	0
Rock Sole	1.2	1	0.26	0	0
Sablefish	9.0	1	0.003	0	0.07
Starry Flounder	4.2	1	0.041	0	0
Black Rockfish	1.8	1	0.18	0	0.05
Boccacio Rockfish	0.0	1	0.92	0	0
Canary Rockfish	0.0	1	0.83	0	0
China Rockfish	0.6	1	0.44	0	0
Copper Rockfish	11.3	1	0.0008	0	0.02
Dusky Rockfish	0.0	1	0.88	0	0
Quillback Rockfish	1.7	1	0.19	0.4	0.2
Redband Rockfish	0.2	1	0.69	0	0
Redstripe Rockfish	0.0	1	0.95	0	0
Tiger Rockfish	4.4	1	0.037	0	0
Widow Rockfish	0.0	1	0.98	0	0
Yelloweye Rockfish	0.9	1	0.35	0.2	0.1
Yellowtail Rockfish	3.6	1	0.06	0	0

Table 11. Average number of harvested fish (and for Chinook Salmon, released) per boat trip, from census of Area 3 lodges, 2017.

Species	June	July	August	Total
Chinook Salmon:				
retained	1.4	0.8	0.2	0.8
released	0.09	0.05	0.03	0.06
Coho Salmon	0.2	2.8	3.9	2.2
Pink Salmon	0	0.01	0.009	0.007
Chum Salmon	0	0.04	0.02	0.02
Sockeye Salmon	0	0.002	0	0.001
Pacific Halibut	0.4	0.6	0.5	0.5
Lingcod	0.3	0.3	0.3	0.3
Other groundfish	0.002	0	0	0.001
Yelloweye Rockfish	0.05	0.07	0.05	0.06
Other rockfish	0.2	0.3	0.4	0.3

Angler Effort Estimates

Over the three month study period, a total of 23 effort surveys were conducted, including 15 weekday surveys and 8 conducted on weekends/holidays (Table 2, Appendix 3). The numbers of lodge boats that were visible during the overflights (Appendix 3) were partitioned among the subareas in Management Area 3, and subtracted from the overflight counts of small boats. The remaining small boat counts ("the non-lodge small boat counts") were significantly correlated to trailer counts at both survey launch sites (Rushbrooke: $F_{1,19} = 62.6$, $P < 0.0001$; Port Edwards: $F_{1,20} = 97.8$, $P < 0.0001$), and when the two launch sites were combined ($F_{1,18} = 184.6$, $P < 0.0001$). The fit of the individual-sites regressions (Rushbrooke $R^2 = 0.75$; Port Edward $R^2 = 0.82$) were lower than that of the combined-site regression ($R^2 = 0.91$; Figure 6).

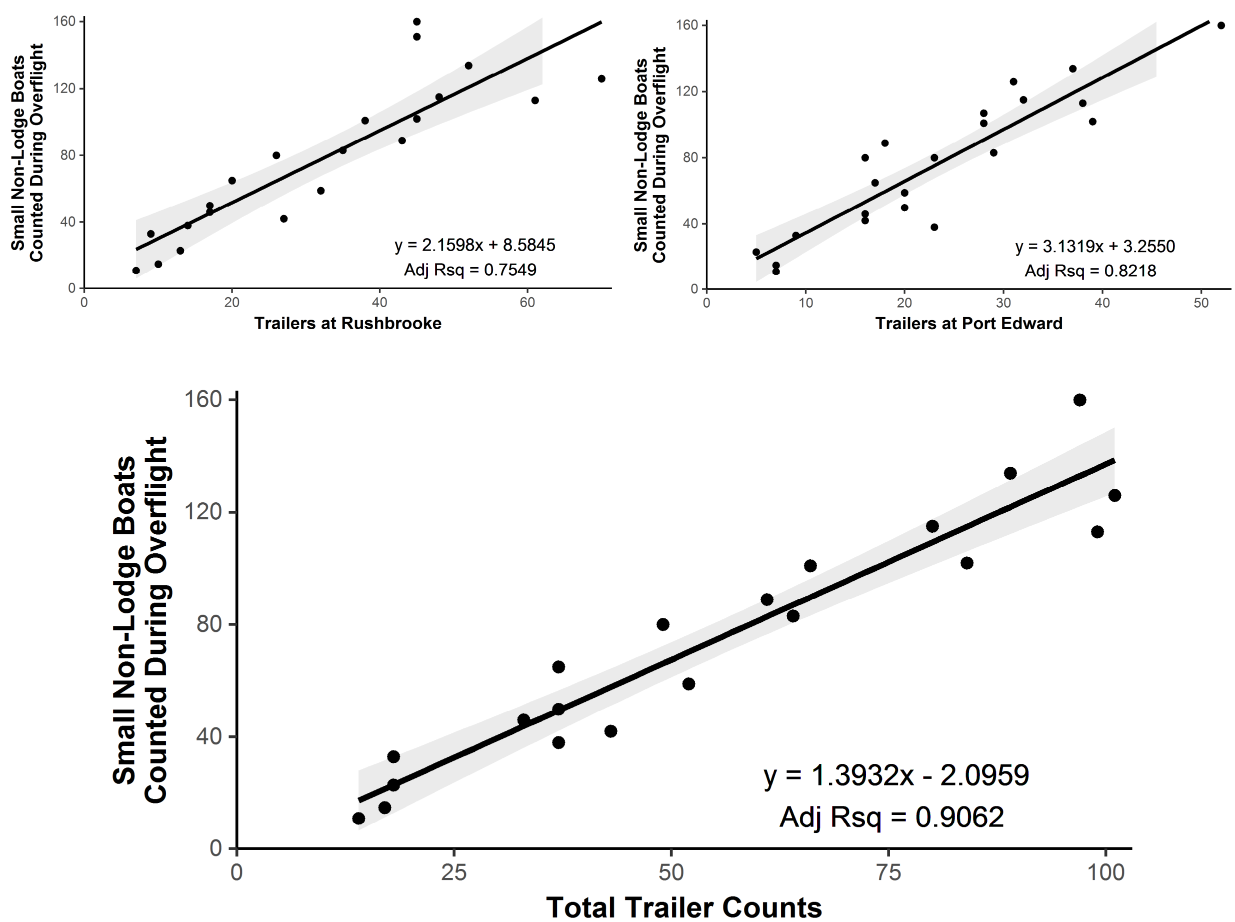

Figure 6. **Linear regressions showing the relationships between small non-lodge boats counted during overflights and trailers concurrently counted at landing sites. Shaded area shows the 95% confidence interval of the predicted values. Note that X axis scale varies among plots.**

The non-lodge boat counts were adjusted for non-lodge angler effort patterns in order to estimate the total monthly non-lodge angling effort for each subarea in 2017 (Table 12, Figures 7 and 8). Median daily non-lodge effort was 8.5 trips per day per subarea on weekdays, and was 10.8 on weekends/holidays, but the difference was not statistically significant ($\chi^2 = 1.6$, df = 1, $P = 0.21$). There was a statistically significant effect of month, whereby effort values in June (medians = 10.6 trips per day per subarea) and July (10.5) were significantly higher than in August (6.7; $\chi^2 = 9.4$, df = 2, $P = 0.009$).

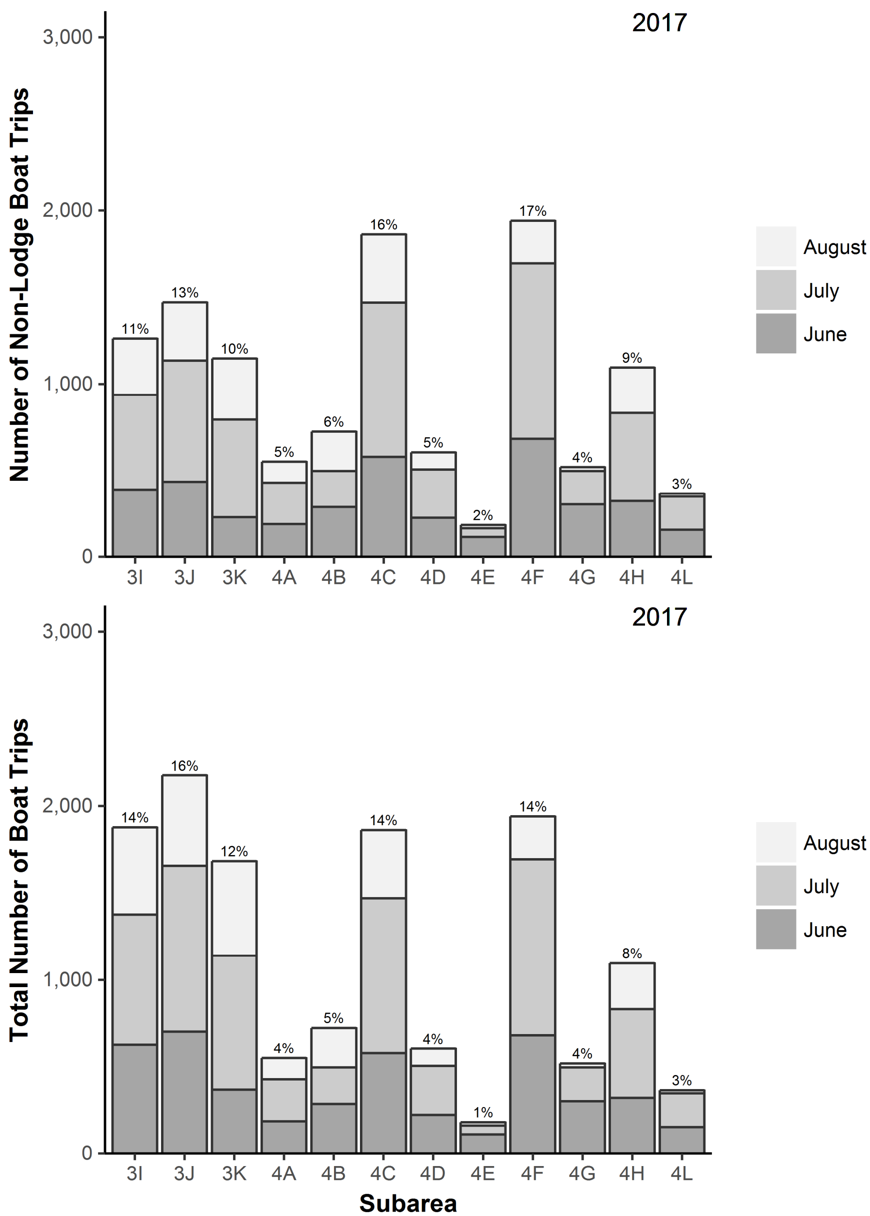

Figure 7. Total estimated numbers of boat trips, by month and subarea, 2017. Upper panel: estimated monthly non-lodge boat trips. Lower panel: total effort, calculated as the sum of lodge and non-lodge boat trips.

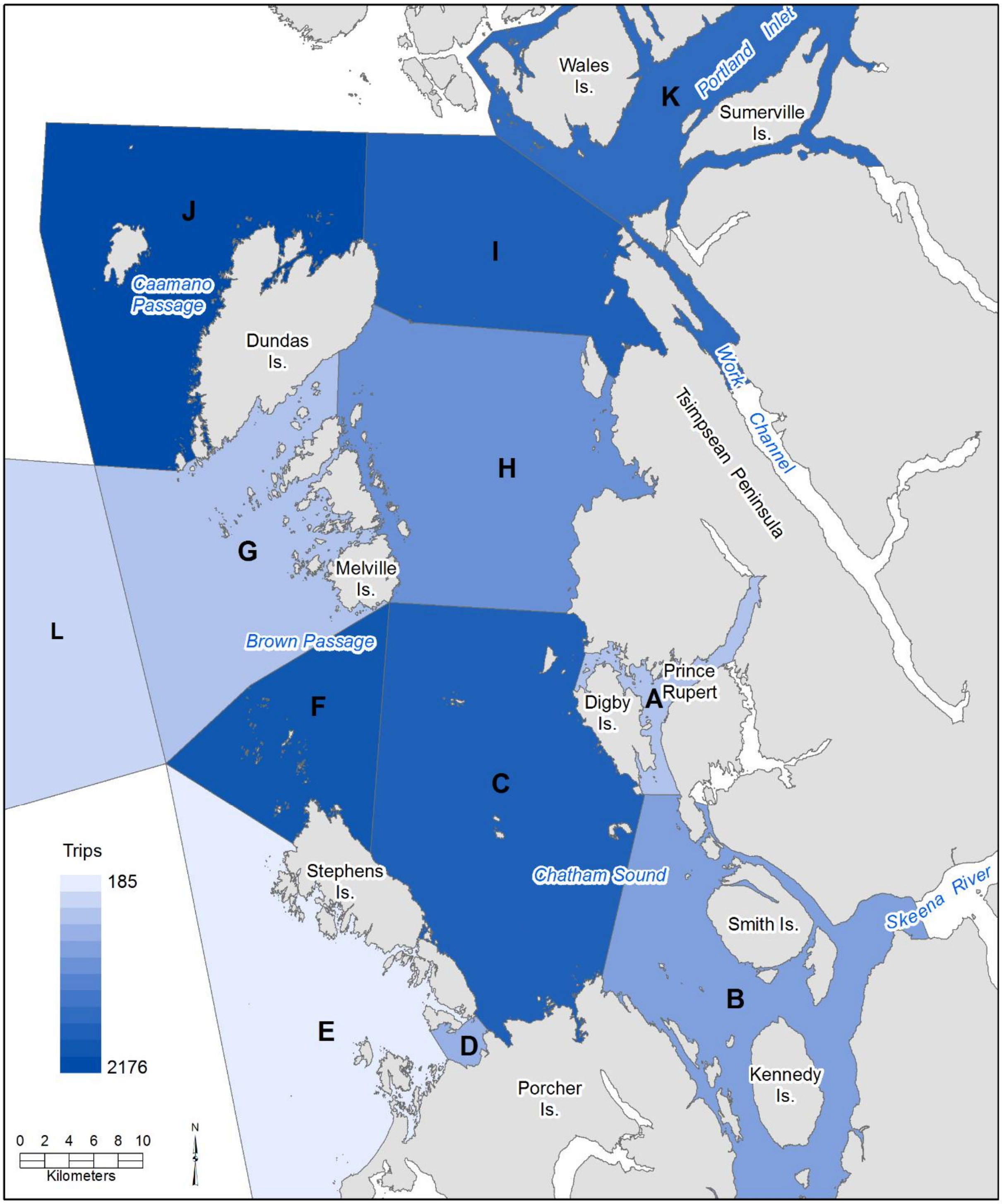

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Figure 8. Heat map showing the estimated total effort (numbers of boat trips, lodge + non-lodge) in each subarea in 2017. See Table 12 for details.

Table 12. Lodge, non-lodge and total effort estimates (boat trips per month), by month and subarea, 2017.

		Subarea												TOTALS			SE of the Totals		
Month		3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June	Creel Estimate	390.1	435.8	229.3	189.1	289.2	579.1	227.0	115.6	683.7	304.7	324.5	156.7	1,055	2,870	3,925	8	12	14
	Lodge Report	239.9	268.0	141.0										649	0	649			
	TOTAL JUNE	630.1	703.8	370.3	189.1	289.2	579.1	227.0	115.6	683.7	304.7	324.5	156.7	1,704	2,870	4,574	8	12	14
	SE	5.0	5.3	2.7	1.5	3.1	4.1	2.8	1.2	7.6	4.1	4.8	2.2						
July	Creel Estimate	547.4	699.6	565.4	240.1	208.3	891.7	278.9	49.2	1013.0	193.7	508.8	192.6	1,812	3,576	5,389	11	12	16
	Lodge Report	197.8	252.8	204.3										655	0	655			
	TOTAL JULY	745.2	952.4	769.7	240.1	208.3	891.7	278.9	49.2	1013.0	193.7	508.8	192.6	2,467	3,576	6,044	11	12	16
	SE	5.0	6.7	6.5	1.7	1.9	5.0	2.7	0.5	8.5	1.8	4.7	2.7						
August	Creel Estimate	325.1	335.8	352.8	122.3	227.6	392.8	100.0	20.2	245.3	22.0	262.0	18.3	1,014	1,411	2,424	7	7	10
	Lodge Report	178.3	184.2	193.5										556	0	556			
	TOTAL AUGUST	503.4	519.9	546.4	122.3	227.6	392.8	100.0	20.2	245.3	22.0	262.0	18.3	1,570	1,411	2,980	7	7	10
	SE	3.6	3.9	4.3	1.2	2.9	3.0	1.4	0.5	3.1	0.6	4.3	0.5						
FULL SEASON TOTAL		1,879	2,176	1,686	551	725	1,864	606	185	1,942	520	1,095	368	5,741	7,856	13,598	15	18	24
FULL SEASON SE		8	9	8	3	5	7	4	1	12	5	8	4						

Total lodge effort in 2017 was 649, 655, and 556 boat trips in June, July and August, respectively (Table 12). To calculate total (lodge + non-lodge) fishing effort by subarea, we distributed the lodge boat trips among the three subareas in Management Area 3, according to the relative non-lodge effort in each (Table 12; Figure 7). In all, there were 3,925 (SE = 14) boat trips in June, 5,389 (SE = 16) boat trips in July, and 2,424 (SE = 10) boat trips in August. The full-season Area 3/4 effort estimate for 2017 was 13,598 boat trips (SE = 24).

Catch Estimates

Monthly and full-season estimates were generated of the numbers of fish released and retained, by species (Tables 13-22; Appendix 4). Non-lodge estimates were generated from the product of effort and CPE estimates. Lodge catches were extracted from census documents. The results presented in this section are total (lodge + non-lodge) numbers of fish retained and released.

An estimated 10,108 (SE = 271) Chinook Salmon were retained by anglers in Areas 3 or 4 during 2017 the study period (Table 14; Figure 9). The largest proportional harvest came from subarea 3J (21%), followed by 4F (18%), and 3I and 3K (15% each). Harvest declined over the study period: it was significantly higher in June (52% of harvest) and July (38% of harvest) than in August (11% of harvest; $\chi^2 = 15.8$, df = 2, $P = 0.0004$; Table 23). During the study, the number of released Chinook Salmon was estimated at 5,308 (SE = 283), or 34% of the total catch.

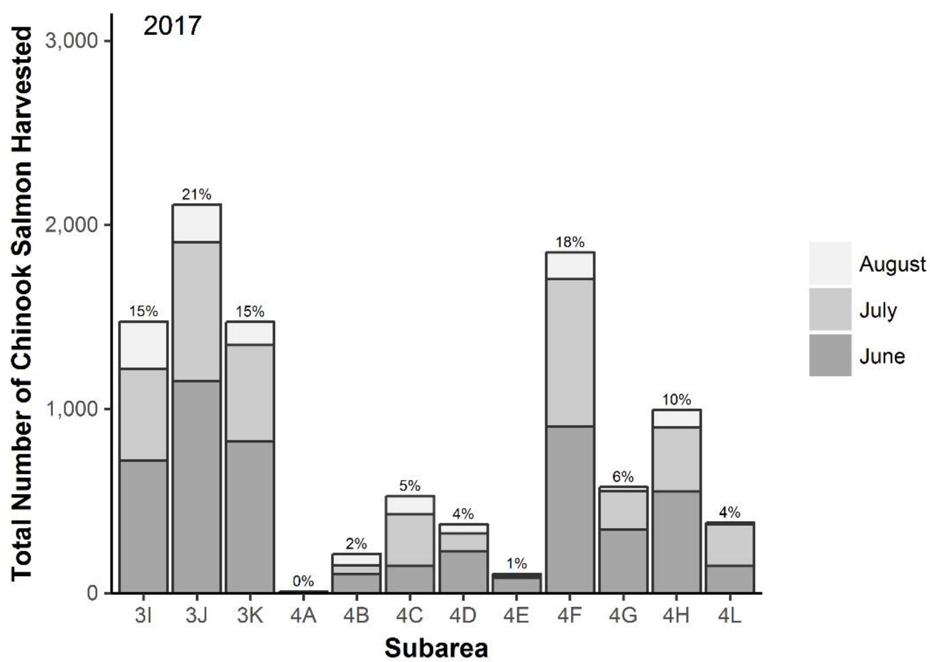

Figure 9. Total (lodge + non-lodge) estimated Chinook Salmon harvest, by month and subarea, 2017.

Table 13. Total (lodge + non-lodge) estimated harvest, by month and subarea, for popular sportfish taxa, 2017.

	Subarea												TOTALS			SE of the Totals		
Species	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
Chinook Salmon	1,476	2,112	1,475	10	213	529	375	107	1,852	577	996	385	5,064	5,044	10,108	197	186	271
Coho Salmon	6,221	6,210	9,465	1	697	2,766	1,003	253	6,287	1,096	3,454	1,260	21,896	16,817	38,713	741	484	885
Pink Salmon	134	79	328	2	24	212	87	7	166	39	149	0	541	686	1,227	78	52	94
Chum Salmon	0	0	57	0	0	3	0	0	27	14	12	13	57	69	127	12	15	20
Sockeye Salmon	37	9	20	1	3	0	3	0	14	21	0	0	66	42	109	28	14	32
Unknown Salmon	0	37	91	0	0	25	0	0	0	0	4	0	129	29	157	47	19	51
Pacific Halibut	1,862	2,932	1,188	2	370	1,537	402	211	2,733	690	1,409	1,269	5,981	8,624	14,606	276	212	348
Lingcod	274	999	0	0	45	169	44	85	513	159	127	358	1,272	1,499	2,772	154	115	193
other groundfish *	179	490	240	11	130	457	71	102	501	34	86	33	908	1,425	2,333	224	115	252
Rockfish †	973	1,714	600	11	285	868	534	148	2,221	391	643	997	3,288	6,101	9,389	266	294	397

* includes Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and unknown groundfish
† includes Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail and unknown rockfish

Table 14. Total (lodge + non-lodge) estimated Chinook Salmon catch, by month and subarea, shown separately for released and retained fish, 2017.

	Subarea												TOTALS			SE of the Totals		
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	722.2	1151.7	824.9	4.8	106.7	150.1	229.2	83.5	906.1	347.2	552.4	150.1	2,699	2,530	5,229	136	124	184
Released	601.4	451.2	643.0	2.9	56.2	96.5	80.1	12.8	412.2	240.9	170.0	32.6	1,696	1,104	2,800	185	113	217
July																		
Retained	497.5	754.2	525.3	3.1	46.3	280.5	96.0	11.6	800.0	209.3	350.5	226.1	1,777	2,023	3,800	109	130	170
Released	262.4	277.2	173.2	1.0	27.0	209.3	9.0	23.1	359.8	212.4	214.8	41.9	713	1,098	1,811	106	104	149
August																		
Retained	256.6	206.1	125.0	2.2	60.3	98.2	50.0	11.6	146.0	20.8	93.3	8.3	588	491	1,078	93	46	104
Released	169.4	192.5	106.1	0	35.6	47.4	13.8	2.1	20.4	30.6	75.4	3.3	468	229	697	99	36	106
TOTAL																		
Retained	1,476	2,112	1,475	10	213	529	375	107	1,852	577	996	385	5,064	5,044	10,108	197	186	271
Released	1,033	921	922	4	119	353	103	38	792	484	460	78	2,876	2,431	5,308	235	158	283
Catch	2,510	3,033	2,397	14	332	882	478	145	2,645	1,061	1,456	462	7,940	7,476	15,416	307	244	392

Table 15. Total (lodge + non-lodge) estimated Coho Salmon catch, by month and subarea, shown separately for released and retained fish, 2017.

	Subarea												TOTALS			SE of the Totals		
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	316.0	450.7	46.4	0	16.8	60.8	53.4	6.4	451.4	106.3	173.8	156.7	813	1,026	1,839	101	98	141
Released	60.7	83.5	0	0	0	14.3	8.9	0	32.7	21.3	30.9	26.1	144	134	278	53	30	61
July																		
Retained	3031.4	2893.4	5740.8	0	194.8	1770.8	602.7	156.3	4606.1	787.2	2023.9	1063.6	11,666	11,205	22,871	500	417	652
Released	357.0	488.4	628.2	0	17.4	414.4	134.9	2.9	951.5	156.2	147.0	41.9	1,474	1,866	3,340	292	217	364
August																		
Retained	2874.1	2865.6	3677.9	1.1	485.3	934.6	346.6	90.2	1229.4	202.9	1256.1	40.0	9,418	4,586	14,004	537	226	583
Released	162.5	251.8	217.1	0	54.8	98.2	17.2	11.6	204.4	15.9	79.0	0	631	481	1,113	137	58	149
TOTAL																		
Retained	6,221	6,210	9,465	1	697	2,766	1,003	253	6,287	1,096	3,454	1,260	21,896	16,817	38,713	741	484	885
Released	580	824	845	0	72	527	161	15	1,189	193	257	68	2,249	2,482	4,731	327	226	398
Catch	6,802	7,033	10,310	1	769	3,293	1,164	267	7,476	1,290	3,711	1,328	24,146	19,299	43,444	810	535	971

Table 16. Total (lodge + non-lodge) estimated Pink Salmon catch, by month and subarea, shown separately for released and retained fish, 2017.

	Subarea												TOTALS			SE of the Totals		
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	0	0	0	0	0	0	4.5	0	0	0	3.9	0	0	8	8	0	4	4
Released	8.7	9.3	0	0	0	0	0	0	6.5	0	0	0	18	7	24	9	3	10
July																		
Retained	134.1	27.0	160.9	2.1	13.5	171.6	57.0	2.9	127.8	37.5	101.8	0	322	514	836	59	46	75
Released	178.5	52.8	274.8	0	9.6	33.5	93.0	0	217.8	0	101.8	16.7	506	472	978	127	64	142
August																		
Retained	0	51.6	166.9	0	11.0	40.6	25.9	3.9	38.0	1.2	43.1	0	218	164	382	51	24	57
Released	185.7	151.1	135.7	0	24.7	40.6	17.2	2.1	14.6	12.2	46.7	8.3	473	166	639	137	32	140
TOTAL																		
Retained	134	79	328	2	24	212	87	7	166	39	149	0	541	686	1,227	78	52	94
Released	373	213	411	0	34	74	110	2	239	12	148	25	997	645	1,642	187	72	200
Catch	507	292	738	2	59	286	197	9	405	51	297	25	1,537	1,331	2,869	203	89	221

Table 17. Total (lodge + non-lodge) estimated Chum Salmon catch, by month and subarea, shown separately for released and retained fish, 2017.

	Subarea												TOTALS			SE of the Totals		
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	0	0	0	0	0	0	0	0	13.1	14.2	11.6	13.1	0	52	52	0	13	13
Released	17.3	0	0	0	0	0	0	0	0	0	0	0	17	0	17	13	0	13
July																		
Retained	0	0	40.1	0	0	0	0	0	4.7	0	0	0	40	5	45	11	4	12
Released	0	0	0	0	0	0	3.0	0	9.5	0	0	0	0	12	12	0	7	7
August																		
Retained	0	0	17.3	0	0	3.4	0	0.4	8.8	0	0	0	17	13	30	5	7	9
Released	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL																		
Retained	0	0	57	0	0	3	0	0	27	14	12	13	57	69	127	12	15	20
Released	17	0	0	0	0	0	3	0	9	0	0	0	17	12	30	13	7	15
Catch	17	0	57	0	0	3	3	0	36	14	12	13	75	82	156	18	17	24

Table 18. Total (lodge + non-lodge) estimated Sockeye Salmon catch, by month and subarea, shown separately for released and retained fish, 2017.

	Subarea												TOTALS			SE of the Totals		
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	0	9.3	0	0	0	0	0	0	0	21.3	0	0	9	21	31	7	11	13
Released	0	0	0	0	0	0	0	0	0	0	7.7	0	0	8	8	0	6	6
July																		
Retained	36.7	0	0	0	0	0	3.0	0	14.2	0	0	0	37	17	54	25	8	27
Released	0	0	0	0	1.9	4.2	0	0	0	0	0	0	0	6	6	0	3	3
August																		
Retained	0	0	20.4	1.1	2.7	0	0	0	0	0	0	0	20	4	24	11	2	11
Released	0	0	0	0	0	3.4	0	0	0	0	0	0	0	3	3	0	3	3
TOTAL																		
Retained	37	9	20	1	3	0	3	0	14	21	0	0	66	42	109	28	14	32
Released	0	0	0	0	2	8	0	0	0	0	8	0	0	17	17	0	7	7
Catch	37	9	20	1	5	8	3	0	14	21	8	0	66	60	126	28	16	33

Table 19. Total (lodge + non-lodge) estimated Pacific Halibut catch, by month and subarea, shown separately for released and retained fish, 2017.

	Subarea												TOTALS			SE of the Totals		
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	471.9	826.0	141.8	1.9	117.9	321.7	180.3	141.3	987.9	425.1	448.1	568.0	1,440	3,192	4,632	100	115	152
Released	424.8	695.4	114.7	2.9	30.9	135.8	95.7	186.3	608.4	184.2	181.6	411.3	1,235	1,837	3,072	259	176	313
July																		
Retained	716.1	1436.0	636.2	0	98.4	653.1	155.9	43.4	1339.7	212.4	627.5	644.8	2,788	3,775	6,563	174	153	231
Released	166.6	726.0	392.6	1.0	13.5	125.6	33.0	17.4	908.9	65.6	254.4	720.2	1,285	2,140	3,425	202	321	380
August																		
Retained	673.6	669.7	410.1	0	153.6	562.1	65.5	26.6	405.9	52.6	333.8	56.7	1,753	1,657	3,410	190	92	211
Released	0	167.9	47.5	0	21.9	125.3	43.1	9.3	116.8	0	118.4	36.7	215	472	687	81	62	102
TOTAL																		
Retained	1,862	2,932	1,188	2	370	1,537	402	211	2,733	690	1,409	1,269	5,981	8,624	14,606	276	212	348
Released	591	1,589	555	4	66	387	172	213	1,634	250	554	1,168	2,735	4,448	7,184	338	372	502
Catch	2,453	4,521	1,743	6	436	1,924	574	424	4,368	940	1,964	2,438	8,717	13,072	21,789	436	428	611

Table 20. Total (lodge + non-lodge) estimated Lingcod catch, by month and subarea, shown separately for released and retained fish, 2017.

	Subarea												TOTALS			SE of the Totals		
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	155.7	266.4	0	0	22.5	50.0	26.7	25.7	206.1	127.5	34.8	208.9	422	702	1,124	61	79	100
Released	34.7	102.0	0	0	5.6	14.3	0	0	62.2	0	0	0	137	82	219	69	21	72
July																		
Retained	118.1	449.0	0	0	5.8	92.1	9.0	55.0	236.7	25.0	56.5	125.6	567	606	1,173	125	78	147
Released	0	0	0	0	1.9	50.2	9.0	20.3	28.4	25.0	22.6	0	0	157	157	0	36	36
August																		
Retained	0	283.3	0	0	16.5	27.1	8.6	4.0	70.1	6.1	35.9	23.3	283	192	475	67	32	74
Released	0	0	0	0	5.5	33.9	0	1.0	20.4	2.4	14.4	0	0	78	78	0	33	33
TOTAL																		
Retained	274	999	0	0	45	169	44	85	513	159	127	358	1,272	1,499	2,772	154	115	193
Released	35	102	0	0	13	98	9	21	111	27	37	0	137	317	454	69	53	87
Catch	308	1,101	0	0	58	268	53	106	624	186	164	358	1,409	1,817	3,226	169	127	211

Table 21. Total (lodge + non-lodge) estimated catch (retained and released) of other groundfish, by month and subarea, shown separately for released and retained fish, 2017. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.

	Subarea												TOTALS			SE of the Totals		
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	95.4	158.6	19.1	1.0	42.1	135.8	53.4	64.2	238.8	21.3	27.0	0	273	584	857	81	69	107
Released	138.7	398.7	219.8	3.8	266.8	986.6	607.6	218.4	1890.7	311.8	216.3	39.2	757	4,541	5,298	205	442	487
July																		
Retained	83.3	264.0	125.6	10.3	65.6	192.6	12.0	34.7	227.2	12.5	22.6	33.5	473	611	1,084	198	78	213
Released	499.8	541.2	400.5	54.6	206.3	1469.4	446.8	159.2	1510.1	218.7	474.9	0	1,441	4,540	5,981	301	436	530
August																		
Retained	0	67.2	95.0	0	21.9	128.7	5.2	3.5	35.0	0	35.9	0	162	230	392	65	48	81
Released	46.4	0	47.5	0	153.6	132.1	37.9	11.6	175.2	74.6	179.4	0	94	764	858	45	136	143
TOTAL																		
Retained	179	490	240	11	130	457	71	102	501	34	86	33	908	1,425	2,333	224	115	252
Released	685	940	668	58	627	2,588	1,092	389	3,576	605	871	39	2,293	9,846	12,138	367	635	734
Catch	864	1,430	907	70	756	3,045	1,163	492	4,077	639	956	73	3,201	11,270	14,471	430	646	776

Table 22. Total (lodge + non-lodge) estimated rockfish catch (retained and released), by month and subarea, shown separately for released and retained fish, 2017. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

	Subarea												TOTALS			SE of the Totals		
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Area 3	Area 4	Overall	Area 3	Area 4	Overall
June																		
Retained	230.6	255.1	0	8.6	75.8	232.4	162.5	77.1	811.2	290.5	220.2	404.8	486	2,283	2,769	60	170	180
Released	121.4	92.7	38.2	6.7	44.9	103.7	55.6	115.6	297.7	120.5	119.8	26.1	252	891	1,143	67	115	133
July																		
Retained	453.6	1107.9	277.0	1.0	121.5	422.8	302.9	52.1	1150.3	87.5	294.0	552.7	1,838	2,985	4,823	219	226	314
Released	83.3	26.4	31.4	2.1	9.6	125.6	102.0	5.8	236.7	34.4	79.1	0	141	595	736	43	91	101
August																		
Retained	289.2	351.1	323.2	1.1	87.7	213.3	69.0	19.3	259.9	13.4	129.2	40.0	964	833	1,796	139	83	162
Released	0	67.2	13.6	5.6	87.7	44.0	36.2	9.8	134.3	0	21.5	0	81	339	420	41	60	72
TOTAL																		
Retained	973	1,714	600	11	285	868	534	148	2,221	391	643	997	3,288	6,101	9,389	266	294	397
Released	205	186	83	14	142	273	194	131	669	155	220	26	474	1,825	2,299	89	158	182
Catch	1,178	1,900	683	25	427	1,142	728	280	2,890	546	864	1,024	3,762	7,926	11,688	281	334	436

Table 23. Kruskal Wallis tests for differences in harvest (fish caught per boat trip per subarea) among months, by species, 2017. Subareas treated as replicates.

	Kruskal-Wallis Statisitics			Median Monthly Harvest		
Species	χ^2	df	P	June	July	August
Chinook Salmon	15.8	2	0.0004	133.9	124.8	32.2
Chum Salmon	2.2	2	0.34	0	0	0
Coho Salmon	17.7	2	0.0001	37.0	690.1	256.1
Pink Salmon	31.3	2	< 0.0001	0	26.0	5.1
Sockeye Salmon	0.5	2	0.79	0	0	0
Arrow Flounder	0.9	2	0.65	0	0	0
Big Skate	11.9	2	0.003	0	0	0
Cabezon	4.1	2	0.13	0	0	0
English Sole	8.3	2	0.015	0	0	0
Greenling	1.5	2	0.48	0	0.2	0.05
Lingcod	8.1	2	0.017	25.0	20.0	4.1
Pacific Cod	5.5	2	0.06	2.1	1.5	0
Pacific Halibut	4.1	2	0.13	156.5	207.4	87.7
Ratfish	2.1	2	0.35	0	0	0
Rock Sole	7.2	2	0.027	0	0	0
Sablefish	2.1	2	0.35	10.6	9.9	3.1
Starry Flounder	1.2	2	0.54	0	0	0
Black Rockfish	7.0	2	0.030	1.0	8.1	3.6
Boccacio Rockfish	6.6	2	0.037	0	0	0
Canary Rockfish	2.0	2	0.36	0	0	0
China Rockfish	4.5	2	0.10	0	0	0
Copper Rockfish	0.2	2	0.89	2.9	0.9	1.4
Dusky Rockfish	6.1	2	0.048	0	0	0
Quillback Rockfish	13.7	2	0.001	18.6	62.7	5.5
Redband Rockfish	7.4	2	0.025	0	0	0
Redstripe Rockfish	7.4	2	0.025	0	0	0
Tiger Rockfish	1.1	2	0.57	0	0	0
Widow Rockfish	4.1	2	0.13	0	0	0
Yelloweye Rockfish	2.0	2	0.37	16.1	17.4	7.5
Yellowtail Rockfish	16.3	2	0.0003	0	0	0.9

The species that was harvested in the greatest numbers during the 2017 study period was, by far, Coho Salmon (38,713 fish retained, SE = 885; Table 15; Figure 10). The largest proportional harvest came from subarea 3K (24%), followed by subareas 3I, 3J and 4F (16% each). In June, 5% of the total Coho Salmon harvest was recorded, significantly less than in July (59% of the total harvest) or August (36% of the total harvest; χ^2 = 17.7, df = 2, *P* = 0.001; Table 23). During the study period, an estimated 4,731 (SE = 398) Coho Salmon were released by anglers (11% of the total catch).

Pink salmon was the least likely salmonid to be retained in 2017 (Table 16). We estimate that 2,869 (SE = 221) fish were caught, of which 43% were retained (1,227 fish, SE = 94), and 57% were released (1,642 fish, SE = 200). Harvest of Pink Salmon varied significantly among months (χ^2 = 31.3, df = 2, *P* < 0.0001; Table 23), with all significant differences for all pairwise comparisons: harvest in July (68% of total) was significantly higher than in August (31% of total), which was significantly higher than in June (1% of total). Very few Chum (127 fish, SE = 20) or Sockeye (109 fish, SE = 32) salmon were harvested in Area 3 and 4 during the study period (Tables 17 and 18).

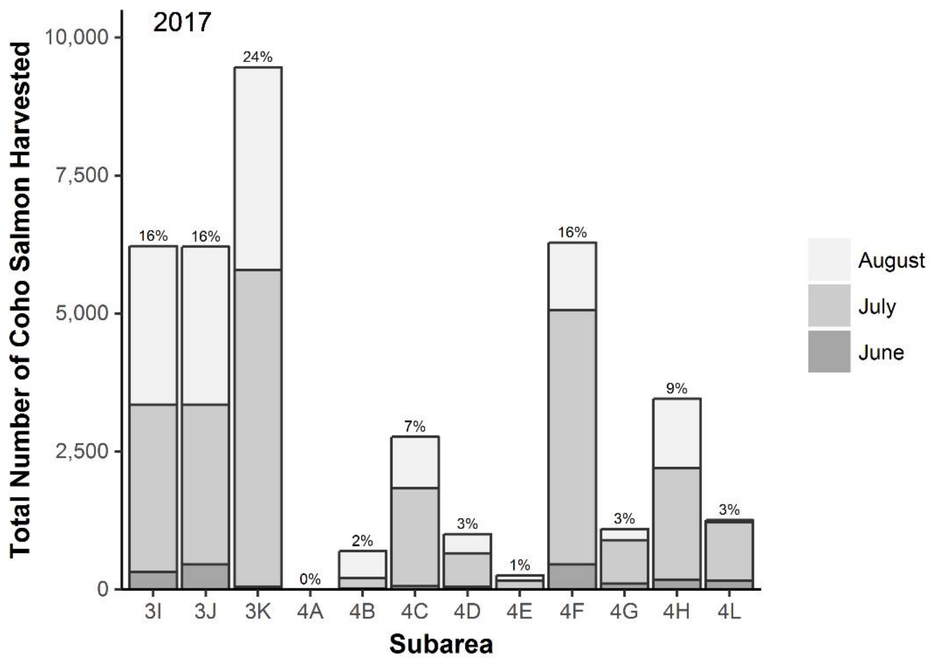

Figure 10. Total (lodge + non-lodge) estimated Coho Salmon harvest, by month and subarea, 2017.

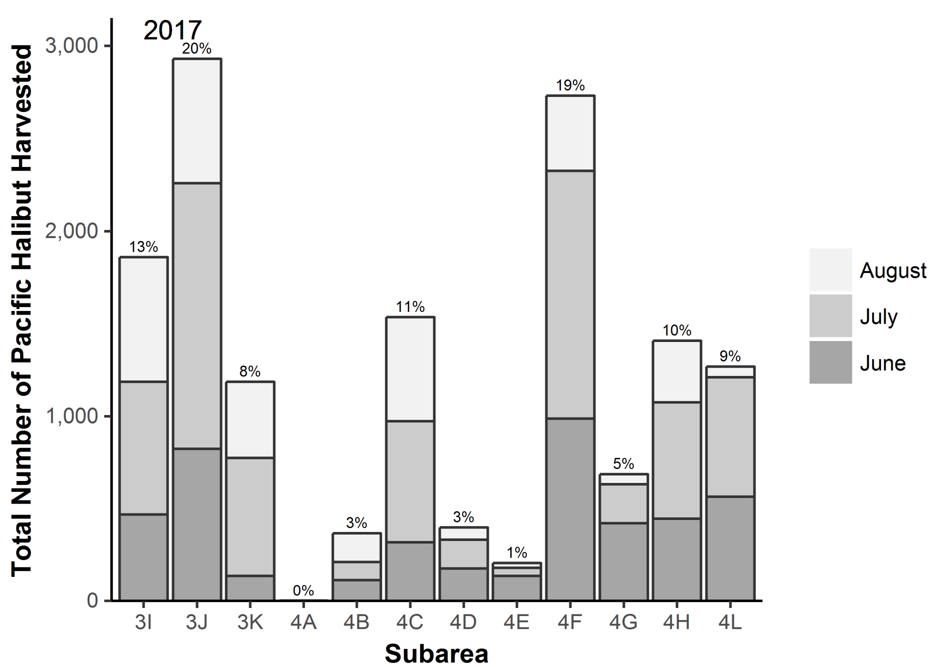

Figure 11. Total (lodge + non-lodge) estimated Pacific Halibut harvest, by month and subarea, 2017.

An estimated 14,606 (SE = 348) Pacific Halibut were harvested in Areas 3 or 4 during the 2017 study period (Table 19; Figure 11). Most of the harvest occurred in subarea 3J (30%) and 4F (19%), followed by 3I (13%). Harvest varied among months, with 32% coming in June, peaking at 45% in July , and returning to 23% in August, though the differences were not statistically significant (χ^2 = 4.1, df = 2, *P* = 0.13; Table 23). We estimated that 7,184 (SE = 502) Pacific Halibut were released (33% of the total catch).

From June to August 2017, an estimated 2,772 (SE = 193) Lingcod were harvested, with significantly more (41-42%) taken in June and July, as compared to August (17%; χ^2 = 8.1, df = 2, *P* = 0.017; Table 20; Figure 12). The harvest occurred largely in subarea 3J (36%), 4F (19%), and 4L (13%). Of the captured Lingcod, 14% were released (454 fish, SE = 87).

An estimated 14,471 (SE = 776) other ground fish were caught June to August 2017, or which 84% was released after capture. During the study period, an estimated 2,333 (SE = 252) individuals were retained (Table 21), mostly (65%) Sablefish (*Anoplopoma fimbria*), but also including Pacific cod (*Gadus macrocephalus*, 17%), greenlings (Hexagrammidae, 7%), Ratfish (*Hydrolagus colliei*), Cabezon (*Scorpaenichthys marmoratus*), and various species of skates, flounders, and soles.

Rockfish harvest was estimated at 9,389 fish (SE = 397; Table 22), composed mostly of Quillback (*Sebastes maliger*, 37%), unknown (22%), Yelloweye (*S. ruberrimus*, 19%), Black (*S. melanops*, 12%), Copper (*S. caurinus*, 3%), China (*S. nebulosus*, 2%), Canary Rockfish (*S. pinniger*, 1%) and seven other rockfish species. In all, 2,299 (SE = 182) were released. Various rockfishes showed significant harvest differences among months (Table 23), most notably for Quillback (harvest in June and July were significantly greater than in August) and Black (harvest in June was significantly less than in July) rockfish.

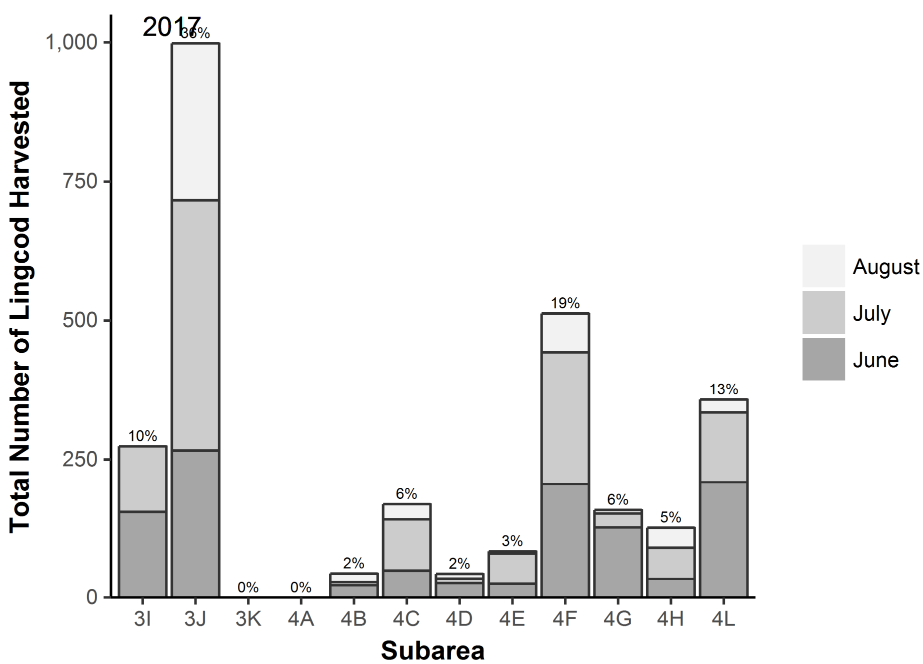

Figure 12. Total (lodge + non-lodge) estimated Lingcod harvest, by month and subarea, 2017.

Biosampling

During interviews, observers were told about 1,797 retained Chinook Salmon, and were permitted to biosample 865 fish. The observed sex ratio was 50:50 (male:female, n = 206), and most fish were red-fleshed (88%; Table 24). The median fork length of biosampled Chinook Salmon (n = 602) was 69 cm (95% of observations were between 46 and 96 cm; Figure 13). Of the biosampled Chinook Salmon, 14% were adipose-clipped (Appendix 5), of which 79% had their heads collected and labelled for submission to the Salmon Head Recovery Program. Scale samples were obtained from 65% of the biosampled Chinook Salmon (Table 24).

Table 24. Biosampling results, 2017.

	Chinook Salmon		Coho Salmon		Pacific Halibut	
	Number	Percent	Number	Percent	Number	Percent
Number Examined	865		1918		1020	
Subarea						
3I	58	7%	70	4%	35	4%
3J	52	6%	61	3%	64	7%
3K	48	6%	227	12%	26	3%
4A	9	1%			2	0%
4B	38	5%	85	5%	64	7%
4C	78	9%	241	13%	170	18%
4D	122	15%	179	10%	101	11%
4E	9	1%	27	1%	22	2%
4F	246	29%	484	26%	278	29%
4G	68	8%	149	8%	45	5%
4H	85	10%	254	14%	105	11%
4L	21	3%	52	3%	42	4%
Sex						
Male	102	50%	259	51%	3	38%
Female	104	50%	245	49%	5	63%
Undeterminable	618		1376		1008	
Unknown	41		38		4	
Trip Type						
Guided	368	43%	868	45%	341	33%
Unguided	497	57%	1050	55%	679	67%
Adipose Clips						
Clipped	120	14%	21	1%		
Unmarked	744	86%	1889	99%		
Unknown	1		8			
Head Tagged						
Yes	95	11%	10	1%		
No	770	89%	1908	99%		
Scale Samples						
Yes	558	65%				
No	307	35%				
Flesh Colour						
Red	352	88%				
White	41	10%				
Marble	9	2%				
Unknown						

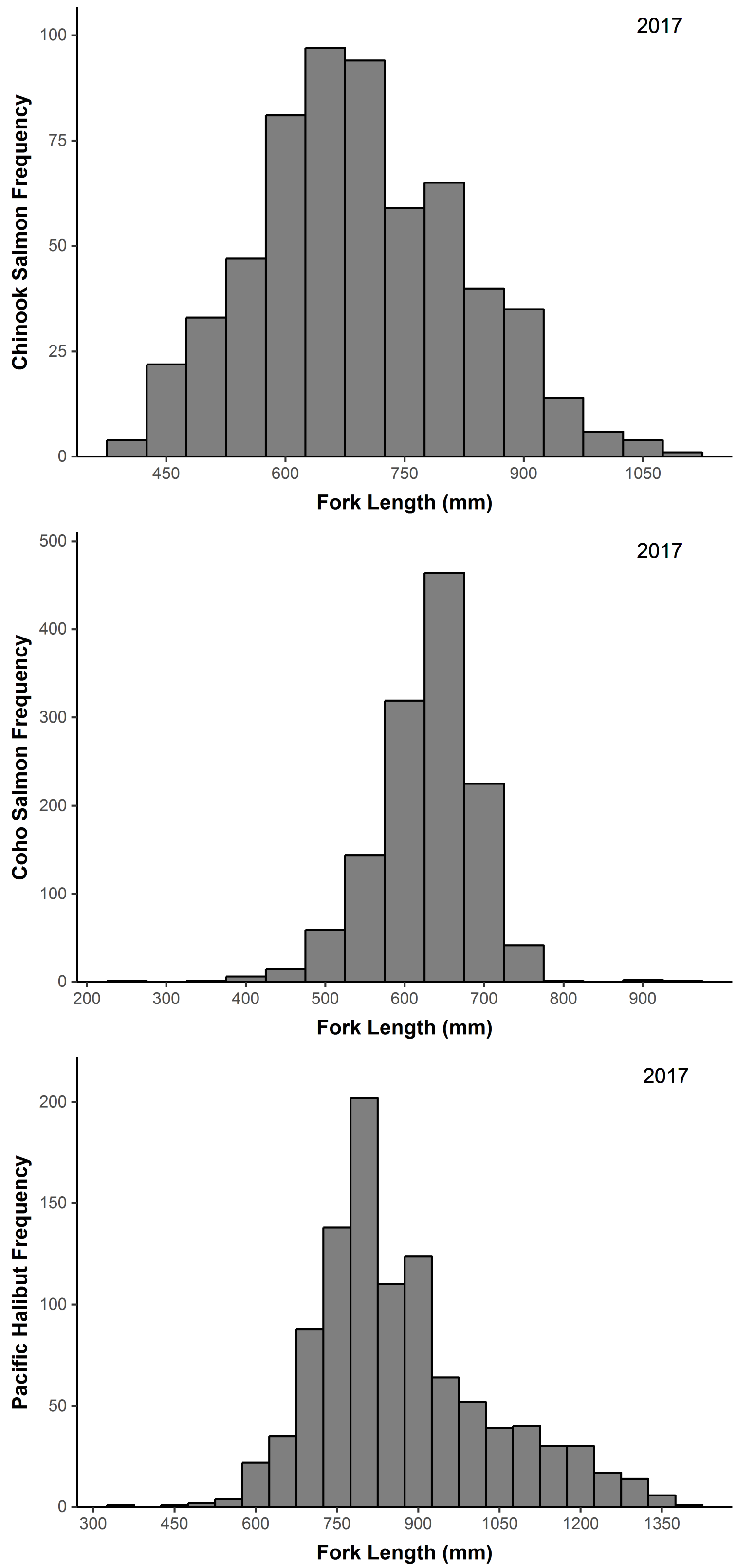

Figure 13. Length-frequency distributions of biosampled Chinook (n = 602) and Coho (n = 1280) salmon, and Pacific Halibut (n = 1020), 2017. Note that X and Y axis scales vary between plots.

For Coho Salmon, observers were told about 6,608 retained fish, and were permitted to biosample 1,918 fish. The observed sex ratio was 51:49 (male:female; n = 504; Table 24). The median Coho Salmon fork length (n = 1,280) was 67 cm (95% of observations were between 48 and 74 cm; Figure 13). Of the biosampled Coho Salmon, 1.1% were adipose-clipped (Appendix 5) of which 48% had their heads were collected and labelled for submission to the Salmon Head Recovery Program).

Of the 2,915 Pacific Halibut that anglers reported as retained, 1020 were biosampled (Table 24). Only eight fish were sexed, and 1020 were measured. The median size was 83 cm (95% of observations were between 61 and 127 cm; Figure 13). Average net and round weights, calculated from average lengths using regression formulas, were 7.1 kg (15.6 lb) and 9.3 kg (20.6 lb) in Area 3, and 6.9 kg (15.2 lb) and 9.1 kg (20.1 lb) in Area 4, respectively (Table 25).

Harvested Halibut Biomass

Based on the average weights of biosampled Pacific Halibut, and the estimated harvest, it was estimated that total harvested Pacific Halibut net biomass was 101.5 tonnes (223,819 lb), including 42.2 t (93,077 lb) in Area 3 and 59.3 t (130,742 lb) in Area 4. In terms of round weight, Pacific Halibut harvest was 134.5 t (296,421 lb) including 55.9 t (123,269 lb) in Area 3 and 78.5 t (173,151 lb) in Area 4 (Table 25).

DISCUSSION

Creel Estimates

Recreational fishing effort, salmon CPE, and salmon catch in Areas 3 & 4 have been increasing over the last 20 years (Table 26). Using published creel data collected since 1995, we found nine years for which effort data were collected in June, July and August (see Van Tongeren and Winther 2010, Robichaud et al. 2016, 2017), and limited analyses to those three months (1995,

Table 25. Average weight, and harvest (numbers and biomass) of Pacific Halibut in Area 3 and 4, 2017.

	Stat Area 3				*Stat Area 4*			
	Net Weight		*Round Weight*		*Net Weight*		*Round Weight*	
	kg	*lb*	*kg*	*lb*	*kg*	*lb*	*kg*	*lb*
Creel Interview (Non-Lodge) Average Weights								
Guided	8.0	17.6	10.6	23.3	7.7	17.0	10.2	22.5
Unguided	6.2	13.7	8.2	18.2	6.5	14.3	8.6	19.0
Overall	7.1	15.6	9.3	20.6	6.9	15.2	9.1	20.1
Estimated Non-lodge Harvest (number of fish)								
Overall	5,023				8,624			
Estimated Lodge Harvest (number of fish)								
Overall	958				0			
Biomass Harvested								
Non-lodge	35,457	78,170	46,959	103,526	59,304	130,742	78,540	173,151
Lodge	6,762	14,907	8,955	19,743	0	0	0	0
Overall	42,219	93,077	55,914	123,269	59,304	130,742	78,540	173,151

1996, 2001, 2002, 2008, 2009, 2015, 2016, 2017 [1]). Meta-analysis revealed (Figure 14) that fishing effort has increased steadily (by 277%) from 4,911 trips in 1995 to 13,598 boat trips in 2017 (an average increase of 335 trips per year). Over the same period, Chinook Salmon harvest has increased six-fold from 1,657 in 1995 to > 10,000 since 2015 (an average increase of 350 Chinook Salmon per year; Table 27, Figure 14). Given that harvest has increased more rapidly than effort, it follows that CPE must have also been increasing over the past 20 years. CPE is measured in 'catch per boat trip', thus CPE can increase if there are more anglers per boat, if more hours are spent fishing per boat trips, if more efficient equipment is used, or if fish abundances increase such that they are caught more readily. It is outside the scope of this report to determine whether the current salmon harvest levels are appropriate for our present-day stock abundances.

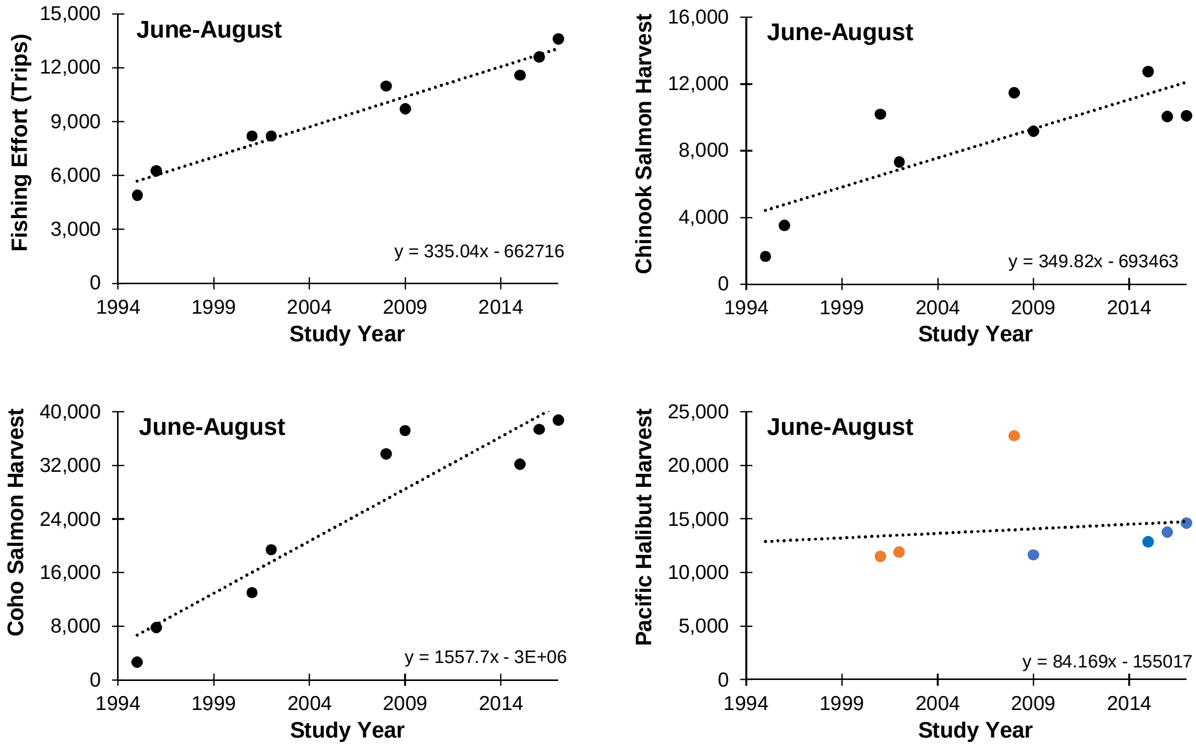

Figure 14. Fishing effort, and harvest of Chinook and Coho salmon, and Pacific Halibut from 1995 to 2017. Annual estimates only include data from June, July and August. Only years with data from all three months are included. Linear regression equations are shown in the lower right of each panel. For Pacific Halibut, bag limits changed after 2008 from two fish per day (orange dots) to one fish per day (blue dots).

[1] Creel data from 2010-2014 were collected and analyzed by DFO, and preliminary results were presented as part of their post-season review process. We based our meta-analysis on published accounts (written accounts showing methods, results, and discussing the caveats of the results), and hence have not included data from these years.

Table 26. Monthly recreational fishing effort (boat trips) in Area 3 and 4 from 1995 to 2017, as estimated using creel survey methods.

	Year																				
Month	1995	1996	1997	1998	1999	2000	2001	2002	2003	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
May	965	391	--	--	--	--	449	960	--	--	--	983	--	--	--	--	--	--	--	--	--
June	1,590	2,490	--	--	--	--	3,801	3,463	--	--	--	4,696	3,043	--	--	--	--	--	3,424	3,961	4,574
July	1,394	2,385	--	1,307	--	1,842	2,588	2,892	--	--	--	3,750	3,804	--	--	--	--	--	4,279	5,295	6,044
Aug	1,927	1,400	--	1,351	--	1,327	1,808	1,843	--	2,593	--	2,529	2,865	--	--	--	--	--	3,881	3,325	2,980
Sept	864	488	--	504	--	954	540	676	--	758	--	394	817	--	--	--	--	--	--	--	--
TOTAL	6,740	7,154	--	3,162	--	4,123	9,186	9,834	--	3,351	--	12,352	10,529	--	--	--	--	--	11,585	12,581	13,598

Table 27. Monthly Chinook Salmon, Coho Salmon and Pacific Halibut harvest in the Area 3 and 4 recreational fishery from 1995 to 2017, as estimated using creel survey methods.

	Year																				
Month	1995	1996	1997	1998	1999	2000	2001	2002	2003	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
Chinook Salmon Harvest																					
May	141	45	--	--	--	--	509	522	--	--	--	474	--	--	--	--	--	--	--	--	--
June	907	2,165	--	--	--	--	6,630	4,028	--	--	--	7,708	4,797	--	--	--	--	--	7,071	7,071	5,229
July	505	1,097	--	254	--	1,611	3,260	2,822	--	--	--	3,080	3,873	--	--	--	--	--	4,016	4,016	3,800
Aug	245	257	--	35	--	108	299	474	--	1,207	--	688	503	--	--	--	--	--	1,669	1,669	1,078
Sept	15	62	--	6	--	93	61	59	--	203	--	20	4	--	--	--	--	--	--	--	--
TOTAL	1,813	3,626	--	295	--	1,812	10,759	7,905	--	1,410	--	11,970	9,177	--	--	--	--	--	12,756	12,757	10,108
Coho Salmon Harvest																					
May	0	0	--	--	--	--	0	0	--	--	--	0	--	--	--	--	--	--	--	--	--
June	335	474	--	--	--	--	1,771	1,332	--	--	--	3,913	3,686	--	--	--	--	--	4,436	4,436	1,839
July	402	3,565	--	1,937	--	568	4,736	9,798	--	--	--	15,982	17,043	--	--	--	--	--	15,101	15,101	22,871
Aug	1,907	3,771	--	1,500	--	2,125	6,488	8,292	--	15,647	--	13,791	16,459	--	--	--	--	--	12,612	12,612	14,004
Sept	1,115	1,417	--	79	--	790	1,006	1,010	--	3,215	--	733	3,136	--	--	--	--	--	--	--	--
TOTAL	3,759	9,227	--	3,516	--	3,483	14,001	20,432	--	18,862	--	34,419	40,324	--	--	--	--	--	32,148	32,149	38,713
Pacific Halibut Harvest																					
May	--	--	--	--	--	--	281	467	--	--	--	1,435	--	--	--	--	--	--	--	--	--
June	--	--	--	--	--	--	4,872	3,912	--	--	--	8,367	3,029	--	--	--	--	--	3,551	3,551	4,632
July	--	--	--	--	--	3,963	3,535	4,716	--	--	--	8,507	4,466	--	--	--	--	--	4,096	4,096	6,563
Aug	--	--	--	1,801	--	2,530	3,085	3,247	--	8,178	--	5,865	4,142	--	--	--	--	--	5,201	5,201	3,410
Sept	--	--	--	328	--	1,224	222	615	--	1,812	--	774	1,205	--	--	--	--	--	--	--	--
TOTAL	--	--	--	2,129	--	7,717	11,995	12,957	--	9,990	--	24,948	12,842	--	--	--	--	--	12,848	12,849	14,606

For Coho Salmon, Table 27 and Figure 14 show a steep (1464%) increase from 2,644 fish in 1995 to over 30,000 fish, yet the dramatic increase can be partly explained by the severe management restrictions in the mid-1990s. For example, if we had data showing higher harvests from before the management actions were in effect (early 1990s), the slope would be considerably shallower. Indeed, harvest appears to have levelled-off since 2009, but it is difficult to draw firm conclusions without examining the 2010-2014 results.

Pacific Halibut harvest did not show a strong among-year trend (Figure 14). While fishing effort increased, the total harvest of Pacific Halibut in recent years (2009, 2015-2017) is very similar to that from 2001-2002; Table 27). This is very likely the result of effective management measures that were enacted between 2008 and 2009 (including the reduction in the daily bag limit from two to one), that appeared to have reduced the catch of halibut by nearly 50% (Figure 14).

The meta-analysis results make two important points: 1) the effort and CPUE of certain species has increased tremendously over the past ~20 years, potentially endangering stocks if left unchecked; and 2) management decisions from the recent past appear to have been effective, indicating that effective tools are available to keep effort and CPUE increases from going unchecked.

Precision of the Results

Overall, our level of survey effort, both for CPE and Effort estimation, were adequate, as evidenced by the relatively good precision of the harvest estimates for most of the taxa surveyed. In 2017 our precision goal was to estimate total harvests within 15% of the true value 19 times out of 20, for the main taxa (Chinook Salmon, Coho Salmon and Pacific Halibut), for which harvest estimates were precise to within 4-5%. Pink Salmon, Lingcod, and rockfish estimates were precise to within 15%, 14%, and 8%, respectively. There were three other groups for which precision was worse than 15%: ‘other groundfish’ was precise to within 21%; and those for Chum and Sockeye salmon were precise to within 30% and 57%, respectively. We did not have pre-defined precision targets for these three groups (the survey was not designed to meet any level of precision for these species).

The precision of our Harvest estimates, depended largely on the variability in the CPE and effort data. Differing levels of angler expertise, and day-to-day changes in abundance of the target species, fishing effort, weather, or fishing conditions mean that fishing trips have a wide range of outcomes. Similarly, day-to-day differences in weather or other factors (such as leisure time, day of week, the price of gasoline, etc.) affect the number of anglers that decide to go out on the water on any given day, resulting in variability within the boat-count dataset. Sampling error is another main source of estimation imprecision; and by meeting our precision goals we have demonstrated that we are sampling an adequate proportion of the population.

Accuracy of the Results

The accuracy the results depends on the quality of data provided by the anglers to the interviewers. Each time an angler is relied upon for unconfirmed data, there is opportunity for results to become inaccurate. For example, in this study, 865 of the 1,797 Chinook Salmon (48%) and 1,918 of the 6,608 Coho Salmon (29%) reported as kept during angler interviews were recorded as observed by the interviewer. While not every fish can be observed[2], each one that goes unconfirmed is nevertheless a possible source of inaccuracy in the reported harvest totals.

In this study, the angler effort estimates depend on the accuracy of the aerial survey boat counts and the activity patterns for each month. For example, we have no estimate of the proportion of angling boats that are missed during overflights, where missed boats would contribute to the underestimation of recreational fishing effort. Also, we do not know if all fishing areas are adequately covered by the current flightpath – perhaps there are fishing locations that are hidden behind islands (e.g., Dunira Island or the Moffat Islands), too far from the flightpath for the surveyors to see active fishing boats. On the other hand, our estimate of fishing effort may be overestimated due to our inability to survey and count boats on foggy or bad weather days. We know that during bad weather, fewer anglers go out on the water; yet these same conditions can cause flights to be cancelled, and thus the lowest effort values are sometimes not sampled.

Another potential source of inaccuracy is the removal of lodge boats from the overflight counts. There is uncertainty around whether all of the active lodge boats were observed by the overflights (e.g. vessels outside of the survey area). There is also uncertainty around the proportion of lodge vessels that were active during the overflights. These factors may influence the accuracy of the estimates of non-lodge fishing effort.

RECOMMENDATIONS

- The catch and effort estimates for the Area 3 and 4 recreational creel surveys for 2010-2014 should be compiled and included in summary tables similar to Table 26 and 27 in this report. If sufficient funding can be obtained, the results from 2010-14 surveys should be compiled into a manuscript report. This additional information will help provide context for the current results, and help clarify how rates of effort and harvest have changed in recent years.

ACKNOWLEDGEMENTS

We would like to thank the creel surveyors (Sara Jordan, Celina Guadagni, Kolby Jones, Renee Samels, Kylie Shepert, and Micheal Standbridge) for collecting large amounts of data and biosamples from anglers. Thanks to Pacific Salmon Commission, who provided support for this work through the Northern Fund. Thanks to Ivan Winther for acting as liaison between our team and DFO. Thanks also go to the administrative staff at LGL and NCSFNSS, whose work made

[2] Fish may not be observed when anglers refuse to provide access to their fish storage containers for various reasons. Reluctance to provide access may be due to fish already being processed and packed in ice, or boat launch traffic congestion causing tension, among other reasons.

it possible for this study to be conducted. We also would like to thank Karl English for providing useful comments and edits on previous drafts of this report. Last but not least, we thank the many anglers, charter operators, and fishing lodges who participated in the survey.

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APPENDICES

APPENDIX 1: DATA FORMS.

2017 AREA 3/4 CREEL SURVEY
AERIAL SURVEY VESSEL ENUMERATION FORM

DATE:____________________ FLIGHT#:_______

SURVEYOR:_____________________________

Note: only survey shaded subareas when conditions permit

Sub	RIGHT FRONT COUNT				TOTAL COUNT				FLIGHT TIME		SEA
Area	SPORT *(Fishing)*	SPORT *(Moving)*	COMM. *(Fishing)*	COMM. *(Moving)*	S *(F)*	S *(M)*	C *(F)*	C *(M)*	START	FINISH	STATE (1-4)
A											
C											
B											
C											
D											
E											
C											
F											
E											
F											
G											
L											
G											
J											
I											
K											
I											
H											
C											
A											

WEATHER / SEA STATE / COMMENTS:

SIGNATURE:______________________________

SHIFT SUMMARY SHEET

Landing Site: ____________________ **Interviewer Name:** ____________________

Interviewer Code: _______________ **Date:** ____________________________

Shift Code: ____________________ **Shift Times:** ____________________________

Time Blocks	
From - To	TB
Before 7:00	1
7:00 - 7:59	2
8:00 - 8:59	3
9:00 - 9:59	4
10:00 - 10:59	5
11:00 - 11:59	6
12:00 - 12:59	7
13:00 - 13:59	8
14:00 - 14:59	9
15:00 - 15:59	10
16:00 - 16:59	11
17:00 - 17:59	12
18:00 - 18:59	13
19:00 - 19:59	14
20:00 - 20:59	15
21:00 - 21:59	16

* Fill in using numbers only.

Time Block	Fishing Conditions	Boats Missed	Not Fishing	Fishing
	Totals:			

Comments: ______________________________________

Trailer Count: __________ **Time:** _________

Database Tally # :	(office use only)

2017 Area 3 and 4 Creel Survey

Site#________ Obs# _______ Date ___/___/___ (DD/MM/YY) Interview# _______

\# of Anglers:______ Guided?:___Y__N Depart. Time:______ Land Time:_______

Pick ONE

- ☐ Refusal ☐ Partial
- ☐ Catch data given by angler (Catch not verified)
- ☐ Fish observed (Catch & Species verified)
- ☐ Adipose checked by interviewer
- ☐ Fish biosampled (e.g., length, sex, scales, etc.)

Pick ONE Target

- ☐ Shellfish Only
- ☐ Finfish Only
- ☐ Finfish and Shellfish

North Coast - Skeena First Nations Stewardship Society

Subarea:________ Total Hours:_________

Species	Kept	Released Legal	Released Non-legal	Gear	# of Lines/ Traps

Target	Hours	Subarea	Site

- ☐ Before 1 AM
- ☐ 0100 - 0159
- ☐ 0200 - 0259
- ☐ 0300 - 0359
- ☐ 0400 - 0459
- ☐ 0500 - 0559
- ☐ 0600 - 0659
- ☐ 0700 - 0759
- ☐ 0800 - 0859
- ☐ 0900 - 0959
- ☐ 1000 - 1059
- ☐ 1100 - 1159
- ☐ 1200 - 1259
- ☐ 1300 - 1359
- ☐ 1400 - 1459
- ☐ 1500 - 1559
- ☐ 1600 - 1659
- ☐ 1700 - 1759
- ☐ 1800 - 1859
- ☐ 1900 - 1959
- ☐ 2000 - 2059
- ☐ 2100 - 2159
- ☐ 2200 - 2259
- ☐ After 11 PM

Subarea:________ Total Hours:_________

Species	Kept	Released Legal	Released Non-legal	Gear	# of Lines/ Traps

Target	Hours	Subarea	Site

- ☐ Before 1 AM
- ☐ 0100 - 0159
- ☐ 0200 - 0259
- ☐ 0300 - 0359
- ☐ 0400 - 0459
- ☐ 0500 - 0559
- ☐ 0600 - 0659
- ☐ 0700 - 0759
- ☐ 0800 - 0859
- ☐ 0900 - 0959
- ☐ 1000 - 1059
- ☐ 1100 - 1159
- ☐ 1200 - 1259
- ☐ 1300 - 1359
- ☐ 1400 - 1459
- ☐ 1500 - 1559
- ☐ 1600 - 1659
- ☐ 1700 - 1759
- ☐ 1800 - 1859
- ☐ 1900 - 1959
- ☐ 2000 - 2059
- ☐ 2100 - 2159
- ☐ 2200 - 2259
- ☐ After 11 PM

Chinook, Coho, and Halibut				Chinook and Coho only		Chinook only		
Species	Sub Area	Fork Length (mm)	Sex N/C=0 Male=1 Female=2 Unable=9	Adipose Clip N=0, Y=1, N/C=9	Head Tag #	Scale Book #	Column # (Fish #)	Flesh Colour R=0, W=1, M=2

Weather Conditions:____

Comments:____

Signature:_

APPENDIX 2: HPE ESTIMATES (AVERAGE NUMBER OF FISH HARVESTED PER BOAT TRIP) BY MONTH, TRIP TYPE, SUBAREA AND SPECIES.

Average number of salmon harvested per boat trip in June 2017, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Chinook Salmon												
Guided	1.40	2.05	3.20		3.83	0.31	2.28	2.00	2.20	1.88	3.00	0.91
Unguided	1.13	1.50	2.16	0.05	0.15	0.26	0.74	0.23	0.84	0.65	0.94	0.86
Combined	1.22	1.74	2.38	0.03	0.37	0.26	1.01	0.72	1.33	1.14	1.70	0.96
Chum Salmon												
Guided									0.05	0.06	0.06	0.04
Unguided										0.04	0.02	0.04
Combined									0.02	0.05	0.04	0.08
Coho Salmon												
Guided	1.33	0.67			0.33	0.31	0.33		1.16	0.71	0.52	1.04
Unguided	0.33	1.00	0.21		0.04	0.09	0.21	0.08	0.38	0.12	0.55	0.40
Combined	0.67	0.85	0.17		0.06	0.10	0.24	0.06	0.66	0.35	0.54	1.00
Pink Salmon												
Guided											0.03	
Unguided							0.02					
Combined							0.02				0.01	
Sockeye Salmon												
Guided		0.05								0.12		
Unguided										0.04		0.01
Combined		0.02								0.07		
Unknown Salmon												
Guided		0.19	0.60									
Unguided												
Combined		0.09	0.13									

Average number of salmon harvested per boat trip in July 2017, by trip type and subarea.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Chinook Salmon												
Guided	1.15	0.75	0.75		0.88	0.50	1.43	1.43	1.46	2.33	0.97	1.93
Unguided	0.45	0.78	0.56	0.02	0.17	0.30	0.32	0.20	0.50	0.78	0.51	
Combined	0.65	0.77	0.67	0.01	0.22	0.31	0.34	0.24	0.79	1.08	0.69	1.17
Chum Salmon												
Guided			0.05				0.01	0.01	0.02			
Unguided												
Combined			0.03						0.00			
Coho Salmon												
Guided	7.31	4.63	11.28		7.00	6.33	8.64	8.64	8.62	6.92	6.17	6.57
Unguided	3.64	3.00	5.19		0.45	1.73	2.07	2.07	2.77	3.38	2.58	3.89
Combined	4.67	3.49	8.57		0.94	1.99	2.16	3.18	4.55	4.06	3.98	5.52
Pink Salmon												
Guided	0.08		0.33		0.38	0.25	0.14	0.14	0.15		0.14	
Unguided	0.30	0.05	0.22	0.01	0.04	0.19	0.21	0.07	0.11	0.24	0.24	
Combined	0.24	0.04	0.28	0.01	0.06	0.19	0.20	0.06	0.13	0.19	0.20	
Sockeye Salmon												
Guided							0.04	0.04	0.05			
Unguided	0.09						0.01					
Combined	0.07						0.01		0.01			
Unknown Salmon												
Guided												
Unguided			0.25			0.03						
Combined			0.11			0.03						

Average number of salmon harvested per boat trip in August 2017, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Chinook Salmon												
Guided	1.00	0.82	0.44		0.33	0.14	1.18	1.18	1.24	0.20	0.41	0.56
Unguided	0.44	0.11	0.15	0.03	0.26	0.26	0.51	0.47	0.38	1.23	0.31	0.52
Combined	0.64	0.50	0.29	0.02	0.27	0.25	0.50	0.58	0.60	0.94	0.36	0.45
Chum Salmon												
Guided			0.04									
Unguided						0.01		0.02	0.05			
Combined			0.02			0.01		0.02	0.04			
Coho Salmon												
Guided	10.00	6.73	9.76		6.50	3.00	6.59	6.59	6.90	12.80	6.97	2.67
Unguided	5.00	6.33	6.37	0.01	1.79	2.34	3.53	4.08	4.38	7.85	2.90	3.98
Combined	6.79	6.55	8.00	0.01	2.13	2.38	3.47	4.46	5.01	9.22	4.79	2.18
Pink Salmon												
Guided		0.27	0.48				0.14	0.14	0.14		0.18	
Unguided			0.44		0.05	0.11	0.26	0.20	0.16	0.08	0.15	0.13
Combined		0.15	0.46		0.05	0.10	0.26	0.19	0.15	0.06	0.16	
Sockeye Salmon												
Guided			0.12									
Unguided				0.01	0.01							
Combined			0.06	0.01	0.01							
Unknown Salmon												
Guided												
Unguided											0.03	0.02
Combined											0.01	

Average number of groundfish harvested per boat trip in June 2017, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Arrow Flounder												
Guided												
Unguided						0.01						
Combined						0.01						
Big Skate												
Guided												
Unguided					0.01				0.01		0.02	0.01
Combined					0.01				0.00		0.01	
Cabezon												
Guided												
Unguided									0.01			
Combined									0.00			
English Sole												
Guided												
Unguided					0.01	0.01						
Combined					0.01	0.01						
Greenling												
Guided									0.04			
Unguided						0.07	0.04				0.02	0.01
Combined						0.06	0.03		0.01		0.01	
Lingcod												
Guided	0.67	0.24			0.50		0.39	0.40	0.33	0.41		1.30
Unguided		0.42			0.05	0.09	0.06	0.15	0.28	0.42	0.17	0.28
Combined	0.22	0.34			0.08	0.09	0.12	0.22	0.30	0.42	0.11	1.33
Pacific Cod												
Guided	0.40				0.33			0.20	0.04		0.10	
Unguided	0.10					0.01	0.01	0.38	0.01	0.08		0.03
Combined	0.20				0.02	0.01	0.01	0.33	0.02	0.05	0.04	
Pacific Halibut												
Guided	0.87	2.05	1.60		1.00	0.69	0.83	1.20	1.92	1.76	1.90	3.61
Unguided	1.03	1.12	0.21	0.02	0.37	0.54	0.79	1.23	1.18	1.15	1.08	1.14
Combined	0.98	1.53	0.50	0.01	0.41	0.56	0.79	1.22	1.44	1.40	1.38	3.63
Ratfish												
Guided												
Unguided									0.01			
Combined									0.00			
Rock Sole												
Guided		0.10							0.01			
Unguided					0.01					0.04		0.01
Combined		0.04			0.01				0.00	0.02		
Pacific Halibut												
Guided	0.87	2.05	1.60		1.00	0.69	0.83	1.20	1.92	1.76	1.90	3.61
Unguided	1.03	1.12	0.21	0.02	0.37	0.54	0.79	1.23	1.18	1.15	1.08	1.14
Combined	0.98	1.53	0.50	0.01	0.41	0.56	0.79	1.22	1.44	1.40	1.38	3.63
Sablefish												
Guided		0.24	0.40		0.17	0.08	0.11		0.23			
Unguided	0.07	0.35			0.08	0.11	0.21	0.23	0.30		0.02	0.01
Combined	0.04	0.30	0.08		0.09	0.10	0.20	0.17	0.27		0.01	
Starry Flounder												
Guided												
Unguided					0.01			0.08				
Combined					0.01			0.06				
Unknown Groundfish												
Guided									0.03			
Unguided		0.04		0.01		0.05			0.01		0.02	0.01
Combined		0.02		0.01		0.04			0.02		0.01	

Average number of groundfish harvested per boat trip in July 2017, by trip type and subarea.

	Subarea											
Species / Trip Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L
Arrow Flounder												
Guided							0.14	0.14	0.15			
Unguided						0.00						
Combined						0.00			0.05			
Big Skate												
Guided												
Unguided			0.06	0.01	0.01	0.00			0.01			
Combined			0.03	0.00	0.01	0.00			0.00			
Cabezon												
Guided												
Unguided												
Combined												
English Sole												
Guided												
Unguided												
Combined												
Greenling												
Guided												0.14
Unguided	0.03			0.01		0.03	0.02		0.05			
Combined	0.02			0.00		0.03	0.02		0.03			0.09
Lingcod												
Guided	0.08	1.13				0.17	0.33	0.33	0.35	0.33	0.11	0.79
Unguided	0.18	0.16			0.03	0.10	0.03	1.27	0.18	0.08	0.11	0.44
Combined	0.15	0.45			0.03	0.10	0.03	1.12	0.23	0.13	0.11	0.65
Pacific Cod												
Guided	0.31						0.06	0.06	0.06		0.09	0.07
Unguided	0.06				0.05	0.01	0.01		0.03			
Combined	0.13				0.05	0.01	0.01		0.04		0.03	0.04
Pacific Halibut												
Guided	1.62	2.75	0.98		1.63	1.58	1.71	1.71	1.82	2.25	2.17	3.14
Unguided	0.94	1.35	0.97		0.38	0.68	0.57	1.00	1.11	0.82	0.64	3.67
Combined	1.13	1.77	0.97		0.47	0.73	0.56	0.88	1.32	1.10	1.23	3.35
Ratfish												
Guided												
Unguided									0.01			
Combined									0.00			
Rock Sole												
Guided												
Unguided				0.04								
Combined				0.03								
Pacific Halibut												
Guided	1.62	2.75	0.98		1.63	1.58	1.71	1.71	1.82	2.25	2.17	3.14
Unguided	0.94	1.35	0.97		0.38	0.68	0.57	1.00	1.11	0.82	0.64	3.67
Combined	1.13	1.77	0.97		0.47	0.73	0.56	0.88	1.32	1.10	1.23	3.35
Sablefish												
Guided												
Unguided		0.54	0.44		0.27	0.17	0.01	0.80	0.13	0.08	0.02	
Combined		0.38	0.19		0.25	0.16	0.01	0.71	0.09	0.06	0.01	
Starry Flounder												
Guided												
Unguided				0.01								
Combined				0.00								
Unknown Groundfish												
Guided												
Unguided					0.01	0.00						0.11
Combined					0.01	0.00						0.04

Average number of groundfish harvested per boat trip in August 2017, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Arrow Flounder												
Guided												
Unguided						0.01					0.05	0.04
Combined						0.01					0.03	
Big Skate												
Guided												
Unguided												
Combined												
Cabezon												
Guided												
Unguided												
Combined												
English Sole												
Guided												
Unguided												
Combined												
Greenling												
Guided			0.04									
Unguided					0.01	0.01		0.02	0.03		0.03	0.02
Combined			0.02		0.01	0.01		0.01	0.02		0.01	
Lingcod												
Guided		0.27			1.00	0.29	0.50	0.50	0.52	1.00		1.33
Unguided		0.56				0.06	0.09	0.15	0.21		0.26	0.22
Combined		0.40			0.07	0.07	0.09	0.20	0.29	0.28	0.14	1.27
Pacific Cod												
Guided			0.12			0.43						
Unguided						0.04		0.04	0.06			
Combined			0.06			0.06		0.03	0.05			
Pacific Halibut												
Guided	1.00	1.73	0.52		2.33	2.71	2.95	2.95	2.86	2.60	1.56	2.89
Unguided	2.11	1.56	1.37		0.55	1.35	0.58	1.02	1.25	2.31	1.03	1.44
Combined	1.71	1.65	0.96		0.67	1.43	0.66	1.32	1.65	2.39	1.27	3.09
Ratfish												
Guided												
Unguided												
Combined												
Rock Sole												
Guided												
Unguided					0.01							
Combined					0.01							
Pacific Halibut												
Guided	1.00	1.73	0.52		2.33	2.71	2.95	2.95	2.86	2.60	1.56	2.89
Unguided	2.11	1.56	1.37		0.55	1.35	0.58	1.02	1.25	2.31	1.03	1.44
Combined	1.71	1.65	0.96		0.67	1.43	0.66	1.32	1.65	2.39	1.27	3.09
Sablefish												
Guided		0.36	0.40			0.29						
Unguided					0.06	0.23	0.05	0.15	0.10		0.18	0.13
Combined		0.20	0.19		0.06	0.23	0.05	0.12	0.07		0.10	
Starry Flounder												
Guided												
Unguided						0.02						
Combined						0.02						
Unknown Groundfish												
Guided					0.17							
Unguided												
Combined					0.01							

Average number of rockfish harvested per boat trip in June 2017, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Black Rockfish												
Guided									0.24			
Unguided	0.03					0.05	0.02	0.15	0.17		0.06	0.04
Combined	0.02					0.04	0.02	0.11	0.20		0.04	
Boccacio Rockfish												
Guided												
Unguided												
Combined												
Canary Rockfish												
Guided												
Unguided										0.08		0.03
Combined										0.05		
China Rockfish												
Guided					0.50							
Unguided						0.01		0.15				
Combined					0.03	0.01		0.11				
Copper Rockfish												
Guided	0.07				0.50				0.04			
Unguided		0.08				0.08	0.02		0.04		0.08	0.05
Combined	0.02	0.04			0.03	0.07	0.02		0.04		0.05	
Dusky Rockfish												
Guided							0.28					
Unguided						0.01					0.06	0.04
Combined						0.01	0.05				0.04	
Quillback Rockfish												
Guided	0.07	0.14			0.50		0.33		0.55	0.53	0.13	0.83
Unguided	0.07			0.02	0.06	0.21	0.51	0.23	0.37	0.54	0.30	0.38
Combined	0.07	0.06		0.01	0.09	0.20	0.48	0.17	0.43	0.53	0.24	0.79
Redband Rockfish												
Guided												
Unguided										0.12		0.04
Combined										0.07		
Redstripe Rockfish												
Guided							0.11					
Unguided	0.07								0.01			
Combined	0.04						0.02		0.00			
Tiger Rockfish												
Guided									0.01			0.04
Unguided									0.01			
Combined									0.01			0.04
Widow Rockfish												
Guided									0.07			
Unguided									0.01			
Combined									0.03			
Yelloweye Rockfish												
Guided	0.33	0.10					0.22	0.20	0.24	0.18	0.03	0.35
Unguided	0.03	0.15			0.03	0.05	0.06	0.31	0.13	0.23	0.21	0.21
Combined	0.13	0.13			0.03	0.05	0.09	0.28	0.17	0.21	0.14	0.33
Yellowtail Rockfish												
Guided												
Unguided				0.01	0.01	0.01			0.01			
Combined				0.01	0.01	0.01			0.01			
Unknown rockfish												
Guided	0.27						0.06		0.25		0.48	1.00
Unguided		0.19		0.06	0.08	0.02	0.04		0.32	0.15		0.19
Combined	0.09	0.11		0.03	0.08	0.02	0.04		0.30	0.09	0.18	1.42

Average number of rockfish harvested per boat trip in July 2017, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Black Rockfish												
Guided	0.54	0.63					0.17	0.17	0.18	0.17		0.07
Unguided	0.15	0.08			0.05	0.08	0.13	0.27	0.12	0.06	0.02	
Combined	0.26	0.25			0.05	0.08	0.13	0.24	0.14	0.08	0.01	0.04
Boccacio Rockfish												
Guided												
Unguided		0.03				0.00				0.02		
Combined		0.02				0.00				0.02		
Canary Rockfish												
Guided							0.01	0.01	0.02			
Unguided						0.02						
Combined						0.02			0.00			
China Rockfish												
Guided		0.06					0.07	0.07	0.08		0.17	
Unguided			0.03						0.02			0.22
Combined		0.02	0.01						0.04		0.07	0.09
Copper Rockfish												
Guided							0.01	0.01	0.02			
Unguided			0.06	0.01	0.08	0.03	0.01		0.05		0.02	
Combined			0.03	0.00	0.07	0.03	0.01		0.04		0.01	
Dusky Rockfish												
Guided							0.06	0.06	0.06			
Unguided						0.02			0.01			
Combined						0.02			0.02			
Quillback Rockfish												
Guided	0.08	0.81	0.43			0.08	0.87	0.87	0.92	0.67	0.46	1.43
Unguided	0.39	0.59	0.25		0.29	0.25	0.31	0.27	0.32	0.16	0.13	0.56
Combined	0.30	0.66	0.35		0.27	0.24	0.30	0.24	0.50	0.26	0.26	1.09
Redband Rockfish												
Guided							0.01	0.01	0.02			0.14
Unguided						0.00						
Combined						0.00			0.00			0.09
Redstripe Rockfish												
Guided												0.36
Unguided												
Combined												0.22
Tiger Rockfish												
Guided							0.03	0.03	0.03			
Unguided												
Combined									0.01			
Widow Rockfish												
Guided												
Unguided												
Combined												
Yelloweye Rockfish												
Guided	0.08	0.13	0.05		0.38	0.08	0.28	0.28	0.29	0.08	0.17	0.57
Unguided	0.09	0.22	0.13			0.06	0.10	0.40	0.21	0.08	0.02	0.11
Combined	0.09	0.19	0.08		0.03	0.06	0.10	0.35	0.23	0.08	0.08	0.39
Yellowtail Rockfish												
Guided												
Unguided										0.02		
Combined										0.02		
Unknown rockfish												
Guided		0.25					0.03	0.03	0.03		0.17	0.86
Unguided	0.09	0.14			0.18	0.01	0.56	0.27	0.19		0.15	1.11
Combined	0.07	0.17			0.17	0.01	0.55	0.24	0.14		0.16	0.96

Average number of rockfish harvested per boat trip in August 2017, by trip type and subarea.

	Subarea											
Species / Trip Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*
Black Rockfish												
Guided			0.08		1.00		0.27	0.27	0.29			
Unguided	0.44				0.09	0.16	0.21	0.25	0.29	0.15	0.03	0.06
Combined	0.29		0.04		0.16	0.15	0.21	0.25	0.29	0.11	0.01	
Boccacio Rockfish												
Guided							0.09	0.09	0.10			
Unguided												
Combined								0.01	0.02			
Canary Rockfish												
Guided							0.09	0.09	0.10			
Unguided											0.38	0.28
Combined								0.01	0.02		0.21	
China Rockfish												
Guided			0.08									0.11
Unguided					0.01							
Combined			0.04		0.01							0.09
Copper Rockfish												
Guided			0.04									
Unguided	0.22				0.04	0.04	0.05	0.05	0.05		0.03	0.02
Combined	0.14		0.02		0.04	0.03	0.05	0.04	0.04		0.01	
Dusky Rockfish												
Guided												
Unguided												
Combined												
Quillback Rockfish												
Guided		0.45	0.24			0.71	0.50	0.50	0.43		0.15	1.00
Unguided		0.33		0.01	0.01	0.13	0.26	0.24	0.17		0.18	0.28
Combined		0.40	0.12	0.01	0.01	0.16	0.29	0.28	0.24		0.16	1.55
Redband Rockfish												
Guided												
Unguided												
Combined												
Redstripe Rockfish												
Guided												
Unguided												
Combined												
Tiger Rockfish												
Guided							0.05	0.05	0.05			0.11
Unguided								0.01	0.02			
Combined								0.01	0.02			0.09
Widow Rockfish												
Guided												
Unguided												
Combined												
Yelloweye Rockfish												
Guided		0.18	0.08		0.67	0.43	0.59	0.59	0.62	1.00	0.03	
Unguided	0.67	1.11			0.04	0.07	0.05	0.20	0.27	0.31	0.10	0.22
Combined	0.43	0.60	0.04		0.08	0.09	0.05	0.26	0.36	0.50	0.07	0.36
Yellowtail Rockfish												
Guided			0.04				0.05	0.05	0.05			
Unguided					0.03	0.06	0.07	0.03			0.05	0.06
Combined			0.02		0.02	0.06	0.07	0.03	0.01		0.03	0.09
Unknown rockfish												
Guided			0.16		0.33							
Unguided					0.04	0.05	0.02	0.05	0.08			
Combined			0.08		0.06	0.04	0.02	0.04	0.06			

APPENDIX 3: COUNTS OF ACTIVELY FISHING SPORT VESSELS IN EACH SUBAREA, BY DATE

Counts of actively fishing sport vessels in each subarea, by date, 2017, along with numbers that were estimated to be lodge boats. Weekend and holiday flights are shaded.

		Subarea												Area 3
Month	Date	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	Lodge Boats
June	1 Jun 2017	2	7	3	1	1	0	1	0	8	0	0	0	11
	5 Jun 2017	3	0	0	2	4	3	8	0	4	0	0	0	0
	8 Jun 2017	0	16	6	5	2	8	1	2	19	2	3	0	12
	13 Jun 2017	7	13	12	6	7	12	4	5	8	11	4	10	12
	17 Jun 2017	13	35	8	4	10	19	8	7	41	23	7	3	11
	18 Jun 2017	10	16	7	6	8	25	13	4	11	10	3	6	11
	25 Jun 2017	23	16	10	3	11	23	8	4	15	9	11	7	8
	28 Jun 2017	28	15	12	4	8	17	0	0	15	4	26	2	8
July	1 Jul 2017	22	29	12	5	2	20	11	1	21	0	6	0	8
	7 Jul 2017	7	10	9	4	9	27	8	2	20	8	5	3	0
	12 Jul 2017	14	21	5	3	4	12	2	0	21	6	11	4	12
	16 Jul 2017	23	24	19	7	6	21	2	0	20	2	17	4	8
	18 Jul 2017	20	20	15	4	3	25	8	2	24	6	23	19	12
	22 Jul 2017	4	17	39	2	2	15	6	1	15	0	3	0	8
	24 Jul 2017	13	8	6	3	6	16	5	1	32	7	11	1	4
	26 Jul 2017	6	14	11	11	2	11	4	1	13	2	8	0	12
August	3 Aug 2017	10	10	17	3	6	14	6	0	9	0	22	2	8
	12 Aug 2017	12	12	16	3	6	7	2	2	7	0	5	0	8
	13 Aug 2017	14	11	13	1	2	8	1	0	5	0	0	0	12
	15 Aug 2017	9	5	8	0	8	12	2	1	3	0	0	0	8
	18 Aug 2017	6	8	6	4	8	3	2	0	8	3	7	0	0
	21 Aug 2017	5	7	3	3	2	10	1	0	2	0	3	0	0
	25 Aug 2017	5	6	5	3	0	1	0	0	1	0	1	0	4

APPENDIX 4: RETAINED AND RELEASED FISH (LODGE REPORTS AND NON-LODGE CREEL ESTIMATES) BY MONTH, SUBAREA AND SPECIES.

Estimated numbers of Chinook Salmon retained by lodge and non-lodge anglers, by month, 2017.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	476.8	760.3	544.6	4.8	106.7	150.1	229.2	83.5	906.1	347.2	552.4	150.1	4,312
Lodge Report	245.4	391.3	280.3	0	0	0	0	0	0	0	0	0	917
JUNE RETAINED	722.2	1151.7	824.9	4.8	106.7	150.1	229.2	83.5	906.1	347.2	552.4	150.1	5,229
SE	52.9	93.2	83.1	1.9	25.3	22.3	23.2	32.6	63.8	58.1	61.7	37.1	184
July													
Creel Estimate	357.0	541.2	376.9	3.1	46.3	280.5	96.0	11.6	800.0	209.3	350.5	226.1	3,298
Lodge Report	140.5	213.1	148.4	0	0	0	0	0	0	0	0	0	502
JULY RETAINED	497.5	754.2	525.3	3.1	46.3	280.5	96.0	11.6	800.0	209.3	350.5	226.1	3,800
SE	50.2	80.7	52.5	1.3	9.3	38.7	12.9	5.4	67.8	40.1	58.0	75.1	170
August													
Creel Estimate	209.0	167.9	101.8	2.2	60.3	98.2	50.0	11.6	146.0	20.8	93.3	8.3	969
Lodge Report	47.6	38.2	23.2	0	0	0	0	0	0	0	0	0	109
AUGUST RETAINED	256.6	206.1	125.0	2.2	60.3	98.2	50.0	11.6	146.0	20.8	93.3	8.3	1,078
SE	72.6	49.9	30.0	1.8	17.5	20.1	10.5	0.4	27.3	6.8	21.9	4.5	104
TOTAL													
Creel Estimate	1,043	1,469	1,023	10	213	529	375	107	1,852	577	996	385	8,580
Lodge Report	434	643	452	0	0	0	0	0	0	0	0	0	1,528
TOTAL RETAINED	1,476	2,112	1,475	10	213	529	375	107	1,852	577	996	385	10,108
SE	103	133	103	3	32	49	29	33	97	71	87	84	271

Estimated numbers of Chinook Salmon released by lodge and non-lodge anglers, by month, 2017.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	580.8	435.8	621.1	2.9	56.2	96.5	80.1	12.8	412.2	240.9	170.0	32.6	2,742
Lodge Report	20.6	15.4	22.0	0	0	0	0	0	0	0	0	0	58
JUNE RELEASED	601.4	451.2	643.0	2.9	56.2	96.5	80.1	12.8	412.2	240.9	170.0	32.6	2,800
SE	104.2	68.9	136.2	1.6	19.6	16.9	17.8	6.3	84.8	53.3	37.4	16.4	217
July													
Creel Estimate	249.9	264.0	164.9	1.0	27.0	209.3	9.0	23.1	359.8	212.4	214.8	41.9	1,777
Lodge Report	12.5	13.2	8.3	0	0	0	0	0	0	0	0	0	34
JULY RELEASED	262.4	277.2	173.2	1.0	27.0	209.3	9.0	23.1	359.8	212.4	214.8	41.9	1,811
SE	71.3	72.1	31.8	0.8	8.5	32.0	3.7	10.7	60.5	43.8	58.1	26.3	149
August													
Creel Estimate	162.5	184.7	101.8	0	35.6	47.4	13.8	2.1	20.4	30.6	75.4	3.3	678
Lodge Report	6.9	7.8	4.3	0	0	0	0	0	0	0	0	0	19
AUGUST RELEASED	169.4	192.5	106.1	0	35.6	47.4	13.8	2.1	20.4	30.6	75.4	3.3	697
SE	81.1	53.1	22.5	0	12.4	16.7	5.2	0.0	7.8	16.3	22.4	3.3	106
TOTAL													
Creel Estimate	993	884	888	4	119	353	103	38	792	484	460	78	5,197
Lodge Report	40	36	35	0	0	0	0	0	0	0	0	0	111
TOTAL RELEASED	1,033	921	922	4	119	353	103	38	792	484	460	78	5,308
SE	150	113	142	2	25	40	19	12	105	71	73	31	283

Estimated numbers of Coho Salmon retained by lodge and non-lodge anglers, by month, 2017.

Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	260.1	370.9	38.2	0	16.8	60.8	53.4	6.4	451.4	106.3	173.8	156.7	1,695
Lodge Report	56.0	79.8	8.2	0	0	0	0	0	0	0	0	0	144
JUNE RETAINED	316.0	450.7	46.4	0	16.8	60.8	53.4	6.4	451.4	106.3	173.8	156.7	1,839
SE	78.0	62.2	16.5	0	6.9	14.2	12.9	4.6	53.9	37.8	35.4	59.4	141
July													
Creel Estimate	2558.4	2442.0	4845.2	0	194.8	1770.8	602.7	156.3	4606.1	787.2	2023.9	1063.6	21,051
Lodge Report	472.9	451.4	895.7	0	0	0	0	0	0	0	0	0	1,820
JULY RETAINED	3031.4	2893.4	5740.8	0	194.8	1770.8	602.7	156.3	4606.1	787.2	2023.9	1063.6	22,871
SE	252.7	329.6	279.0	0	39.1	134.1	72.6	37.8	256.2	85.5	181.0	205.7	652
August													
Creel Estimate	2205.7	2199.2	2822.6	1.1	485.3	934.6	346.6	90.2	1229.4	202.9	1256.1	40.0	11,814
Lodge Report	668.3	666.4	855.3	0	0	0	0	0	0	0	0	0	2,190
AUGUST RETAINED	2874.1	2865.6	3677.9	1.1	485.3	934.6	346.6	90.2	1229.4	202.9	1256.1	40.0	14,004
SE	405.2	299.5	187.1	0.9	80.1	99.4	36.4	1.8	96.0	28.9	150.7	22.2	583
TOTAL													
Creel Estimate	5,024	5,012	7,706	1	697	2,766	1,003	253	6,287	1,096	3,454	1,260	34,559
Lodge Report	1,197	1,198	1,759	0	0	0	0	0	0	0	0	0	4,154
TOTAL RETAINED	6,221	6,210	9,465	1	697	2,766	1,003	253	6,287	1,096	3,454	1,260	38,713
SE	484	450	336	1	89	167	82	38	279	98	238	215	885

Estimated numbers of Coho Salmon released by lodge and non-lodge anglers, by month, 2017.

Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	60.7	83.5	0	0	0	14.3	8.9	0	32.7	21.3	30.9	26.1	278
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	60.7	83.5	0	0	0	14.3	8.9	0	32.7	21.3	30.9	26.1	278
SE	37.9	37.1	0	0	0	8.0	5.0	0	11.4	15.3	16.8	12.9	61
July													
Creel Estimate	357.0	488.4	628.2	0	17.4	414.4	134.9	2.9	951.5	156.2	147.0	41.9	3,340
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	357.0	488.4	628.2	0	17.4	414.4	134.9	2.9	951.5	156.2	147.0	41.9	3,340
SE	95.5	219.7	167.2	0	7.1	109.2	27.4	2.3	167.8	52.6	53.4	21.4	364
August													
Creel Estimate	162.5	251.8	217.1	0	54.8	98.2	17.2	11.6	204.4	15.9	79.0	0	1,113
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	162.5	251.8	217.1	0	54.8	98.2	17.2	11.6	204.4	15.9	79.0	0	1,113
SE	88.3	92.7	48.2	0	18.2	28.7	7.4	0.3	38.7	6.4	25.1	0	149
TOTAL													
Creel Estimate	580	824	845	0	72	527	161	15	1,189	193	257	68	4,731
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	580	824	845	0	72	527	161	15	1,189	193	257	68	4,731
SE	135	241	174	0	20	113	29	2	173	55	61	25	398

Estimated numbers of Pink Salmon retained by lodge and non-lodge anglers, by month, 2017.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	0	0	0	0	0	4.5	0	0	0	3.9	0	8
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	0	0	0	0	0	0	4.5	0	0	0	3.9	0	8
SE	0	0	0	0	0	0	3.2	0	0	0	3.0	0	4
July													
Creel Estimate	130.9	26.4	157.1	2.1	13.5	171.6	57.0	2.9	127.8	37.5	101.8	0	828
Lodge Report	3.2	0.7	3.9	0	0	0	0	0	0	0	0	0	8
JULY RETAINED	134.1	27.0	160.9	2.1	13.5	171.6	57.0	2.9	127.8	37.5	101.8	0	836
SE	40.5	18.7	39.0	1.1	4.4	23.3	14.7	2.3	22.5	15.0	24.0	0	75
August													
Creel Estimate	0	50.4	162.8	0	11.0	40.6	25.9	3.9	38.0	1.2	43.1	0	377
Lodge Report	0.0	1.2	4.0	0	0	0	0	0	0	0	0	0	5
AUGUST RETAINED	0	51.6	166.9	0	11.0	40.6	25.9	3.9	38.0	1.2	43.1	0	382
SE	0	27.5	43.5	0	4.3	13.9	7.8	0.1	8.6	1.2	15.7	0	57
TOTAL													
Creel Estimate	131	77	320	2	24	212	87	7	166	39	149	0	1,214
Lodge Report	3	2	8	0	0	0	0	0	0	0	0	0	13
TOTAL RETAINED	134	79	328	2	24	212	87	7	166	39	149	0	1,227
SE	40	33	58	1	6	27	17	2	24	15	29	0	94

Estimated numbers of Pink Salmon released by lodge and non-lodge anglers, by month, 2017.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	8.7	9.3	0	0	0	0	0	0	6.5	0	0	0	24
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	8.7	9.3	0	0	0	0	0	0	6.5	0	0	0	24
SE	6.3	6.6	0	0	0	0	0	0	3.3	0	0	0	10
July													
Creel Estimate	178.5	52.8	274.8	0	9.6	33.5	93.0	0	217.8	0	101.8	16.7	978
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	178.5	52.8	274.8	0	9.6	33.5	93.0	0	217.8	0	101.8	16.7	978
SE	104.1	26.2	67.7	0	3.9	10.5	24.9	0	44.9	0	33.9	14.9	142
August													
Creel Estimate	185.7	151.1	135.7	0	24.7	40.6	17.2	2.1	14.6	12.2	46.7	8.3	639
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	185.7	151.1	135.7	0	24.7	40.6	17.2	2.1	14.6	12.2	46.7	8.3	639
SE	104.1	78.3	41.6	0	14.1	13.4	7.1	0.0	5.7	12.2	18.9	5.7	140
TOTAL													
Creel Estimate	373	213	411	0	34	74	110	2	239	12	148	25	1,642
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	373	213	411	0	34	74	110	2	239	12	148	25	1,642
SE	147	83	79	0	15	17	26	0	45	12	39	16	200

Estimated numbers of Chum Salmon retained by lodge and non-lodge anglers, by month, 2017.

	Subarea												
Month / Catch Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*	*TOTAL*
June													
Creel Estimate	0	0	0	0	0	0	0	0	13.1	14.2	11.6	13.1	52
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	0	0	0	0	0	0	0	0	13.1	14.2	11.6	13.1	52
SE	0	0	0	0	0	0	0	0	5.7	7.1	6.6	6.4	13
July													
Creel Estimate	0	0	15.7	0	0	0	0	0	4.7	0	0	0	20
Lodge Report	0	0	24.4	0	0	0	0	0	0	0	0	0	24
JULY RETAINED	0	0	40.1	0	0	0	0	0	4.7	0	0	0	45
SE	0	0	11.4	0	0	0	0	0	3.6	0	0	0	12
August													
Creel Estimate	0	0	6.8	0	0	3.4	0	0.4	8.8	0	0	0	19
Lodge Report	0.0	0.0	10.6	0	0	0	0	0	0	0	0	0	11
AUGUST RETAINED	0	0	17.3	0	0	3.4	0	0.4	8.8	0	0	0	30
SE	0	0	4.9	0	0	2.7	0	0.0	6.6	0	0	0	9
TOTAL													
Creel Estimate	0	0	22	0	0	3	0	0	27	14	12	13	92
Lodge Report	0	0	35	0	0	0	0	0	0	0	0	0	35
TOTAL RETAINED	0	0	57	0	0	3	0	0	27	14	12	13	127
SE	0	0	12	0	0	3	0	0	9	7	7	6	20

Estimated numbers of Chum Salmon released by lodge and non-lodge anglers, by month, 2017.

	Subarea												
Month / Catch Type	*3I*	*3J*	*3K*	*4A*	*4B*	*4C*	*4D*	*4E*	*4F*	*4G*	*4H*	*4L*	*TOTAL*
June													
Creel Estimate	17.3	0	0	0	0	0	0	0	0	0	0	0	17
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	17.3	0	0	0	0	0	0	0	0	0	0	0	17
SE	12.5	0	0	0	0	0	0	0	0	0	0	0	13
July													
Creel Estimate	0	0	0	0	0	0	3.0	0	9.5	0	0	0	12
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	0	0	3.0	0	9.5	0	0	0	12
SE	0	0	0	0	0	0	2.2	0	7.1	0	0	0	7
August													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	0	0	0
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL													
Creel Estimate	17	0	0	0	0	0	3	0	9	0	0	0	30
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	17	0	0	0	0	0	3	0	9	0	0	0	30
SE	13	0	0	0	0	0	2	0	7	0	0	0	15

Estimated numbers of Sockeye Salmon retained by lodge and non-lodge anglers, by month, 2017.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	9.3	0	0	0	0	0	0	0	21.3	0	0	31
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	0	9.3	0	0	0	0	0	0	0	21.3	0	0	31
SE	0	6.6	0	0	0	0	0	0	0	11.3	0	0	13
July													
Creel Estimate	35.7	0	0	0	0	0	3.0	0	14.2	0	0	0	53
Lodge Report	1.0	0	0	0	0	0	0	0	0	0	0	0	1
JULY RETAINED	36.7	0	0	0	0	0	3.0	0	14.2	0	0	0	54
SE	25.4	0	0	0	0	0	2.2	0	8.0	0	0	0	27
August													
Creel Estimate	0	0	20.4	1.1	2.7	0	0	0	0	0	0	0	24
Lodge Report	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	0	0	20.4	1.1	2.7	0	0	0	0	0	0	0	24
SE	0	0	10.9	0.9	2.2	0	0	0	0	0	0	0	11
TOTAL													
Creel Estimate	36	9	20	1	3	0	3	0	14	21	0	0	108
Lodge Report	1	0	0	0	0	0	0	0	0	0	0	0	1
TOTAL RETAINED	37	9	20	1	3	0	3	0	14	21	0	0	109
SE	25	7	11	1	2	0	2	0	8	11	0	0	32

Estimated numbers of Sockeye Salmon released by lodge and non-lodge anglers, by month, 2017.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	7.7	0	8
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	0	0	0	0	0	0	0	0	0	7.7	0	8
SE	0	0	0	0	0	0	0	0	0	0	6.0	0	6
July													
Creel Estimate	0	0	0	0	1.9	4.2	0	0	0	0	0	0	6
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	1.9	4.2	0	0	0	0	0	0	6
SE	0	0	0	0	1.5	3.1	0	0	0	0	0	0	3
August													
Creel Estimate	0	0	0	0	0	3.4	0	0	0	0	0	0	3
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	0	0	0	3.4	0	0	0	0	0	0	3
SE	0	0	0	0	0	2.7	0	0	0	0	0	0	3
TOTAL													
Creel Estimate	0	0	0	0	2	8	0	0	0	0	8	0	17
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	0	0	0	0	2	8	0	0	0	0	8	0	17
SE	0	0	0	0	1	4	0	0	0	0	6	0	7

Estimated numbers of unknown salmon retained by lodge and non-lodge anglers, by month, 2017.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	37.1	28.7	0	0	0	0	0	0	0	0	0	66
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RETAINED	0	37.1	28.7	0	0	0	0	0	0	0	0	0	66
SE	0	26.2	20.9	0	0	0	0	0	0	0	0	0	34
July													
Creel Estimate	0	0	62.8	0	0	25.1	0	0	0	0	0	0	88
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	0	0	62.8	0	0	25.1	0	0	0	0	0	0	88
SE	0	0	32.9	0	0	18.4	0	0	0	0	0	0	38
August													
Creel Estimate	0	0	0	0	0	0	0	0	0	0	3.6	0	4
Lodge Report	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	0	0	0	0	0	0	0	0	0	0	3.6	0	4
SE	0	0	0	0	0	0	0	0	0	0	3.2	0	3
TOTAL													
Creel Estimate	0	37	91	0	0	25	0	0	0	0	4	0	157
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RETAINED	0	37	91	0	0	25	0	0	0	0	4	0	157
SE	0	26	39	0	0	18	0	0	0	0	3	0	51

Estimated numbers of unknown salmon released by lodge and non-lodge anglers, by month, 2017.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	0	37.1	28.7	0	0	0	0	0	13.1	0	0	0	79
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	0	37.1	28.7	0	0	0	0	0	13.1	0	0	0	79
SE	0	26.2	20.9	0	0	0	0	0	7.4	0	0	0	34
July													
Creel Estimate	0	0	133.5	0	0	12.6	0	0	4.7	0	0	0	151
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	133.5	0	0	12.6	0	0	4.7	0	0	0	151
SE	0	0	73.5	0	0	9.2	0	0	3.6	0	0	0	74
August													
Creel Estimate	0	0	0	0	2.7	0	0	0	0	0	0	0	3
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	0	0	2.7	0	0	0	0	0	0	0	3
SE	0	0	0	0	2.2	0	0	0	0	0	0	0	2
TOTAL													
Creel Estimate	0	37	162	0	3	13	0	0	18	0	0	0	232
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	0	37	162	0	3	13	0	0	18	0	0	0	232
SE	0	26	76	0	2	9	0	0	8	0	0	0	82

Estimated numbers of Pacific Halibut retained by lodge and non-lodge anglers, by month, 2017.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	381.4	667.6	114.7	1.9	117.9	321.7	180.3	141.3	987.9	425.1	448.1	568.0	4,356
Lodge Report	90.5	158.3	27.2	0	0	0	0	0	0	0	0	0	276
JUNE RETAINED	471.9	826.0	141.8	1.9	117.9	321.7	180.3	141.3	987.9	425.1	448.1	568.0	4,632
SE	51.0	75.6	40.3	1.5	31.4	31.7	21.3	33.5	57.5	55.6	46.6	33.7	152
July													
Creel Estimate	618.8	1240.8	549.7	0	98.4	653.1	155.9	43.4	1339.7	212.4	627.5	644.8	6,184
Lodge Report	97.3	195.2	86.5	0	0	0	0	0	0	0	0	0	379
JULY RETAINED	716.1	1436.0	636.2	0	98.4	653.1	155.9	43.4	1339.7	212.4	627.5	644.8	6,563
SE	92.7	129.1	70.2	0	16.4	54.2	20.8	11.7	85.5	34.6	63.2	83.9	231
August													
Creel Estimate	557.2	554.0	339.3	0	153.6	562.1	65.5	26.6	405.9	52.6	333.8	56.7	3,107
Lodge Report	116.4	115.7	70.9	0	0	0	0	0	0	0	0	0	303
AUGUST RETAINED	673.6	669.7	410.1	0	153.6	562.1	65.5	26.6	405.9	52.6	333.8	56.7	3,410
SE	131.9	124.0	57.0	0	26.7	62.4	12.5	0.7	41.8	12.5	41.9	9.8	211
TOTAL													
Creel Estimate	1,557	2,462	1,004	2	370	1,537	402	211	2,733	690	1,409	1,269	13,648
Lodge Report	304	469	185	0	0	0	0	0	0	0	0	0	958
TOTAL RETAINED	1,862	2,932	1,188	2	370	1,537	402	211	2,733	690	1,409	1,269	14,606
SE	169	194	99	1	44	89	32	35	111	67	89	91	348

Estimated numbers of Pacific Halibut released by lodge and non-lodge anglers, by month, 2017.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	424.8	695.4	114.7	2.9	30.9	135.8	95.7	186.3	608.4	184.2	181.6	411.3	3,072
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	424.8	695.4	114.7	2.9	30.9	135.8	95.7	186.3	608.4	184.2	181.6	411.3	3,072
SE	119.3	218.2	70.6	2.2	18.2	33.0	27.3	83.7	81.3	47.3	49.5	102.6	313
July													
Creel Estimate	166.6	726.0	392.6	1.0	13.5	125.6	33.0	17.4	908.9	65.6	254.4	720.2	3,425
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	166.6	726.0	392.6	1.0	13.5	125.6	33.0	17.4	908.9	65.6	254.4	720.2	3,425
SE	46.7	167.0	103.6	0.8	4.9	26.5	9.2	10.1	209.2	21.8	51.6	235.4	380
August													
Creel Estimate	0	167.9	47.5	0	21.9	125.3	43.1	9.3	116.8	0	118.4	36.7	687
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	167.9	47.5	0	21.9	125.3	43.1	9.3	116.8	0	118.4	36.7	687
SE	0	78.4	21.1	0	11.2	31.5	14.7	0.2	29.3	0	38.4	15.0	102
TOTAL													
Creel Estimate	591	1,589	555	4	66	387	172	213	1,634	250	554	1,168	7,184
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	591	1,589	555	4	66	387	172	213	1,634	250	554	1,168	7,184
SE	128	286	127	2	22	53	32	84	226	52	81	257	502

Estimated numbers of Lingcod retained by lodge and non-lodge anglers, by month, 2017.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	86.7	148.4	0	0	22.5	50.0	26.7	25.7	206.1	127.5	34.8	208.9	937
Lodge Report	69.0	118.0	0	0	0	0	0	0	0	0	0	0	187
JUNE RETAINED	155.7	266.4	0	0	22.5	50.0	26.7	25.7	206.1	127.5	34.8	208.9	1,124
SE	44.6	42.2	0	0	11.6	15.4	10.8	10.7	29.9	36.6	15.9	56.1	100
July													
Creel Estimate	83.3	316.8	0	0	5.8	92.1	9.0	55.0	236.7	25.0	56.5	125.6	1,006
Lodge Report	34.8	132.2	0	0	0	0	0	0	0	0	0	0	167
JULY RETAINED	118.1	449.0	0	0	5.8	92.1	9.0	55.0	236.7	25.0	56.5	125.6	1,173
SE	38.2	118.9	0	0	3.3	25.0	3.7	21.1	43.2	10.7	20.8	50.0	147
August													
Creel Estimate	0	134.3	0	0	16.5	27.1	8.6	4.0	70.1	6.1	35.9	23.3	326
Lodge Report	0.0	149.0	0.0	0	0	0	0	0	0	0	0	0	149
AUGUST RETAINED	0	283.3	0	0	16.5	27.1	8.6	4.0	70.1	6.1	35.9	23.3	475
SE	0	66.7	0	0	13.3	10.6	7.2	0.1	15.8	6.1	18.3	9.0	74
TOTAL													
Creel Estimate	170	599	0	0	45	169	44	85	513	159	127	358	2,269
Lodge Report	104	399	0	0	0	0	0	0	0	0	0	0	503
TOTAL RETAINED	274	999	0	0	45	169	44	85	513	159	127	358	2,772
SE	59	143	0	0	18	31	14	24	55	39	32	76	193

Estimated numbers of Lingcod released by lodge and non-lodge anglers, by month, 2017.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	34.7	102.0	0	0	5.6	14.3	0	0	62.2	0	0	0	219
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	34.7	102.0	0	0	5.6	14.3	0	0	62.2	0	0	0	219
SE	19.6	65.8	0	0	2.8	10.1	0	0	18.4	0	0	0	72
July													
Creel Estimate	0	0	0	0	1.9	50.2	9.0	20.3	28.4	25.0	22.6	0	157
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	0	0	0	0	1.9	50.2	9.0	20.3	28.4	25.0	22.6	0	157
SE	0	0	0	0	1.5	21.6	6.5	12.2	10.0	18.5	13.7	0	36
August													
Creel Estimate	0	0	0	0	5.5	33.9	0	1.0	20.4	2.4	14.4	0	78
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	0	0	0	5.5	33.9	0	1.0	20.4	2.4	14.4	0	78
SE	0	0	0	0	3.1	26.8	0	0.0	13.4	2.4	12.7	0	33
TOTAL													
Creel Estimate	35	102	0	0	13	98	9	21	111	27	37	0	454
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	35	102	0	0	13	98	9	21	111	27	37	0	454
SE	20	66	0	0	4	36	7	12	25	19	19	0	87

Estimated numbers of other groundfish retained by lodge and non-lodge anglers, by month, 2017. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	95.4	157.6	19.1	1.0	42.1	135.8	53.4	64.2	238.8	21.3	27.0	0	856
Lodge Report	0	1.0	0	0	0	0	0	0	0	0	0	0	1
JUNE RETAINED	95.4	158.6	19.1	1.0	42.1	135.8	53.4	64.2	238.8	21.3	27.0	0	857
SE	41.5	68.5	14.0	0.7	10.7	33.4	14.0	25.6	50.9	8.7	8.9	0	107
July													
Creel Estimate	83.3	264.0	125.6	10.3	65.6	192.6	12.0	34.7	227.2	12.5	22.6	33.5	1,084
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RETAINED	83.3	264.0	125.6	10.3	65.6	192.6	12.0	34.7	227.2	12.5	22.6	33.5	1,084
SE	32.2	186.7	57.6	5.5	18.5	40.4	5.3	27.8	52.1	11.8	10.6	18.2	213
August													
Creel Estimate	0	67.2	95.0	0	21.9	128.7	5.2	3.5	35.0	0	35.9	0	392
Lodge Report	0.0	0.0	0.0	0	0	0	0	0	0	0	0	0	0
AUGUST RETAINED	0	67.2	95.0	0	21.9	128.7	5.2	3.5	35.0	0	35.9	0	392
SE	0	50.2	41.4	0	7.6	41.3	4.3	0.9	14.9	0	17.3	0	81
TOTAL													
Creel Estimate	179	489	240	11	130	457	71	102	501	34	86	33	2,332
Lodge Report	0	1	0	0	0	0	0	0	0	0	0	0	1
TOTAL RETAINED	179	490	240	11	130	457	71	102	501	34	86	33	2,333
SE	53	205	72	6	23	67	16	38	74	15	22	18	252

Estimated numbers of other groundfish released by lodge and non-lodge anglers, by month, 2017. Included are Big and Longnose skates, Dogfish, Pacific Cod, Sablefish, Cabezon, Red Irish Lord, Greenling, Ratfish, Arrow and Starry flounders, Pacific Sanddab, Rock and English soles, and other miscellaneous groundfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	138.7	398.7	219.8	3.8	266.8	986.6	607.6	218.4	1890.7	311.8	216.3	39.2	5,298
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	138.7	398.7	219.8	3.8	266.8	986.6	607.6	218.4	1890.7	311.8	216.3	39.2	5,298
SE	56.1	174.0	92.1	1.8	75.6	210.7	105.1	103.8	306.6	158.0	61.8	19.7	487
July													
Creel Estimate	499.8	541.2	400.5	54.6	206.3	1469.4	446.8	159.2	1510.1	218.7	474.9	0	5,981
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	499.8	541.2	400.5	54.6	206.3	1469.4	446.8	159.2	1510.1	218.7	474.9	0	5,981
SE	170.5	180.1	170.3	38.2	59.6	317.4	108.8	80.5	227.5	71.4	95.9	0	530
August													
Creel Estimate	46.4	0	47.5	0	153.6	132.1	37.9	11.6	175.2	74.6	179.4	0	858
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	46.4	0	47.5	0	153.6	132.1	37.9	11.6	175.2	74.6	179.4	0	858
SE	33.7	0	29.9	0	52.9	42.5	13.8	0.3	71.9	73.3	56.8	0	143
TOTAL													
Creel Estimate	685	940	668	58	627	2,588	1,092	389	3,576	605	871	39	12,138
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	685	940	668	58	627	2,588	1,092	389	3,576	605	871	39	12,138
SE	183	250	196	38	110	383	152	131	388	188	127	20	734

Estimated numbers of rockfish retained by lodge and non-lodge anglers, by month, 2017. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	147.4	148.4	0	8.6	75.8	232.4	162.5	77.1	811.2	290.5	220.2	404.8	2,579
Lodge Report	83.2	106.8	0	0	0	0	0	0	0	0	0	0	190
JUNE RETAINED	230.6	255.1	0	8.6	75.8	232.4	162.5	77.1	811.2	290.5	220.2	404.8	2,769
SE	39.8	45.2	0	4.7	21.8	40.3	29.3	19.8	100.6	68.8	55.7	86.9	180
July													
Creel Estimate	392.7	910.8	267.0	1.0	121.5	422.8	302.9	52.1	1150.3	87.5	294.0	552.7	4,555
Lodge Report	60.9	197.1	10.0	0	0	0	0	0	0	0	0	0	268
JULY RETAINED	453.6	1107.9	277.0	1.0	121.5	422.8	302.9	52.1	1150.3	87.5	294.0	552.7	4,823
SE	95.7	181.1	76.5	0.8	35.7	55.4	71.5	15.7	106.8	23.9	60.5	159.8	314
August													
Creel Estimate	278.6	335.8	122.1	1.1	87.7	213.3	69.0	19.3	259.9	13.4	129.2	40.0	1,569
Lodge Report	10.6	15.4	201.0	0	0	0	0	0	0	0	0	0	227
AUGUST RETAINED	289.2	351.1	323.2	1.1	87.7	213.3	69.0	19.3	259.9	13.4	129.2	40.0	1,796
SE	92.5	97.8	33.9	0.9	20.5	37.8	15.5	0.6	40.8	7.1	53.2	13.5	162
TOTAL													
Creel Estimate	819	1,395	389	11	285	868	534	148	2,221	391	643	997	8,704
Lodge Report	155	319	211	0	0	0	0	0	0	0	0	0	685
TOTAL RETAINED	973	1,714	600	11	285	868	534	148	2,221	391	643	997	9,389
SE	139	211	84	5	47	78	79	25	152	73	98	182	397

Estimated numbers of rockfish released by lodge and non-lodge anglers, by month, 2017. Included are Black, Boccacio, Canary, China, Copper, Dusky, Quillback, Redband, Redstripe, Tiger, Widow, Yelloweye, Yellowtail, and other miscellaneous rockfishes.

	Subarea												
Month / Catch Type	3I	3J	3K	4A	4B	4C	4D	4E	4F	4G	4H	4L	TOTAL
June													
Creel Estimate	121.4	92.7	38.2	6.7	44.9	103.7	55.6	115.6	297.7	120.5	119.8	26.1	1,143
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JUNE RELEASED	121.4	92.7	38.2	6.7	44.9	103.7	55.6	115.6	297.7	120.5	119.8	26.1	1,143
SE	54.5	31.6	21.8	3.9	14.7	28.4	15.1	56.7	56.1	53.4	48.3	18.6	133
July													
Creel Estimate	83.3	26.4	31.4	2.1	9.6	125.6	102.0	5.8	236.7	34.4	79.1	0	736
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
JULY RELEASED	83.3	26.4	31.4	2.1	9.6	125.6	102.0	5.8	236.7	34.4	79.1	0	736
SE	34.6	18.7	18.0	1.1	4.9	30.4	26.9	4.6	71.2	20.5	33.5	0	101
August													
Creel Estimate	0	67.2	13.6	5.6	87.7	44.0	36.2	9.8	134.3	0	21.5	0	420
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
AUGUST RELEASED	0	67.2	13.6	5.6	87.7	44.0	36.2	9.8	134.3	0	21.5	0	420
SE	0	39.7	9.8	3.3	46.2	13.3	18.2	0.4	27.0	0	13.5	0	72
TOTAL													
Creel Estimate	205	186	83	14	142	273	194	131	669	155	220	26	2,299
Lodge Report	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL RELEASED	205	186	83	14	142	273	194	131	669	155	220	26	2,299
SE	65	54	30	5	49	44	36	57	95	57	60	19	182

APPENDIX 5: ADIPOSE FIN-CLIP INCIDENCE

Incidence of fin clipping observed in biosamples, by species, month and subarea, in 2017.

Species / Month	Adipose	3I	3J	3K	3?	4A	4B	4C	4D	4E	4F	4G	4H	4L	4?	TOTAL
Chinook Salmon																
June	Clipped	7	1	5	1	0	1	6	3	1	22	4	10	0	1	62
	Not Clipped	37	36	33	8	4	18	31	81	7	153	30	48	5	9	500
	Percent Clipped	16%	3%	13%	11%	0%	5%	16%	4%	13%	13%	12%	17%	0%	10%	11%
July	Clipped	6	0	1	1	0	4	9	4	0	12	10	2	2	0	51
	Not Clipped	8	9	7	4	3	11	25	15	1	49	23	12	14	7	188
	Percent Clipped	43%	0%	13%	20%	0%	27%	26%	21%	0%	20%	30%	14%	13%	0%	21%
August	Clipped	0	1	0	0	0	0	1	2	0	3	0	0	0	0	7
	Not Clipped	0	5	2	0	2	4	5	17	0	7	1	13	0	0	56
	Percent Clipped	-	17%	0%	-	0%	0%	17%	11%	-	30%	0%	0%	-	-	11%
Total	Clipped	13	2	6	2	0	5	16	9	1	37	14	12	2	1	120
	Not Clipped	45	50	42	12	9	33	61	113	8	209	54	73	19	16	744
	Percent Clipped	22%	4%	13%	14%	0%	13%	21%	7%	11%	15%	21%	14%	10%	6%	14%
Coho Salmon																
June	Clipped	0	0	0	0	0	0	0	0	0	3	0	0	0	0	3
	Not Clipped	11	18	1	0	0	5	15	15	1	77	11	10	0	1	165
	Percent Clipped	0%	0%	0%	-	-	0%	0%	0%	0%	4%	0%	0%	-	0%	1.8%
July	Clipped	0	0	2	0	0	0	3	0	1	3	0	2	2	0	13
	Not Clipped	59	15	135	7	0	35	132	68	25	314	95	79	48	35	1,047
	Percent Clipped	0%	0%	1%	0%	-	0%	2%	0%	4%	1%	0%	2%	4%	0%	1.2%
August	Clipped	0	0	0	1	0	0	1	0	0	1	0	2	0	0	5
	Not Clipped	0	28	89	30	0	45	90	96	0	81	43	158	2	15	677
	Percent Clipped	-	0%	0%	3%	-	0%	1%	0%	-	1%	0%	1%	0%	0%	0.7%
Total	Clipped	0	0	2	1	0	0	4	0	1	7	0	4	2	0	21
	Not Clipped	70	61	225	37	0	85	237	179	26	472	149	247	50	51	1,889
	Percent Clipped	0%	0%	1%	3%	-	0%	2%	0%	4%	1%	0%	2%	4%	0%	1.1%